

Applied Microeconomics — Master Notes

by Jakob V. Stangel

One running document for the whole course, organised by lecture (= textbook chapter). Format per lecture: **Résumé** → ★ **Heart of it** → **Key concepts** → **Section-by-section** → **Cases** → **Key terms** → **Exam pointers**. Textbook: Richard Friberg, *A Free Introduction to Microeconomics* · Lecturer: Luigi Butera (lbu.eco@cbs.dk). Exam: written sit-in, *open book, No AI*, Week 50 (9 Dec 2026).

Overview — where everything is

(Everything below is clickable.)

- **Lecture 1: Principles of Economics (Ch. 1)** — what is economics? · the three principles · opportunity cost & marginal thinking · models: positive vs normative · ★ Fun facts
- **Lecture 2: Supply and Demand (Ch. 2)** — demand & the demand function · supply & the supply function · ★ solving market equilibrium · shifts & comparative statics · price ceilings & floors · ★ Fun facts
- **Lecture 3: Elasticities & Taxes (Ch. 2 recap · §6.4)** — why elasticity? · ★ price elasticity of demand · computing & the point-elasticity trick · elastic / inelastic / unit-elastic · cross-price · income elasticity & Engel curves · supply elasticity · elasticity over time · ★ taxes: the wedge, incidence & equivalence · formula sheet · ★ practice problems (with solutions) · ★ Fun facts
- **Lecture 4: Consumer Choice (Ch. 5)** — modelling choice: the 3 ingredients · preferences & rationality axioms · utility & utility functions · ★ marginal utility · indifference curves & their 3 types · ★ marginal rate of substitution (MRS) · budget constraint & MRT · ★ the optimum: $MRS = MRT$ · comparative statics · interior vs corner · Cobb-Douglas shortcut · behavioral economics · cases · formula sheet · ★ practice problems (with solutions) · ★ Fun facts

Lecture 1: Principles of Economics (Chapter 1)

Required reading: Friberg, ch. 1 (*Introduction*).

Theme of the lecture (Butera): the "economic way of thinking" — economics studies how people allocate scarce resources among competing needs, using **three principles** (optimization · equilibrium · empiricism) and **models** (simplified, testable) to understand and predict behaviour.

★ **The heart of the chapter:** economics = doing the best you can with scarce resources. The toolkit is **three principles** — **Optimization** · **Equilibrium** · **Empiricism**; the two workhorses of optimization are **opportunity cost** and **marginal thinking**; and the method is building **positive (testable) models**, not making normative judgments. "*All models are useful, but all models are wrong.*"

The red thread. Everything starts from **scarcity** — without it we wouldn't need economics. From scarcity flows: you must **choose** (optimize) → every choice has an **opportunity cost** and is decided **at the margin** → many people choosing interact until no one wants to move (**equilibrium**) → and we test our claims about all this with **models + data** (empiricism). For any decision ask: *what's scarce · what's the best alternative given up (opportunity cost) · what does one more unit give me (marginal) · is this claim positive (testable) or normative (a value judgment)?*

Key concepts / "modes" to use

- **Scarcity** — limited resources vs unlimited wants; the reason economics exists.
- **The three principles: Optimization · Equilibrium · Empiricism.**
- **Opportunity cost** — the value of the *best alternative* you give up.
- **Marginal thinking** — decide by the value of *one more* unit; **law of diminishing marginal returns.**
- **Model** — a simplified representation of relationships between variables ("all models are wrong, but useful").
- **Positive vs normative** — a *testable* cause-effect claim vs a *value judgment*.

Section-by-section main points

| What is economics?

Core: economics = **the study of how people allocate scarce resources over competing needs** — "people (try to) do the best they can for themselves with what they have."

- **Scarcity of two things:** the **things we want** (goods, services, jobs, education, relationships, recognition...) *and* the **means to get them** (money, **time, attention**, information, patience, willpower). Air to breathe isn't scarce → not an economic question.
- Every decision is shaped by **scarcity** (can't have it all), **other people** (you're not an island), and **the rules of the game** (you can't do whatever you want).
- **How societies allocate: central planners** (someone at the top decides who gets what — parents, government) vs **markets** (value set by supply & demand — from job markets to dating markets). Economics gives tools to predict how we decide and when markets do (or don't) allocate *efficiently*.

| The three principles of economics

Core: the whole course rests on three ideas.

Principle	Meaning
① Optimization	We choose the best feasible option given our limited resources (money, time, attention, willpower are all scarce).
② Equilibrium	A situation in which no one would benefit from changing their <i>own</i> behaviour, given the choices of everyone else.
③ Empiricism	We build models → models generate predictions → we use data/experiments to test (falsify) them.

Optimization — opportunity cost + marginal thinking

Core: to optimise, use two tools — opportunity cost (weigh each use against its best alternative — "apples to apples") and marginal thinking (what does *one more* unit give me?).

Opportunity cost = the value of the best alternative you give up — nothing "free" is free if it uses a scarce resource.

OPPORTUNITY COST

$$OC(\text{action}) = \text{benefit}(\text{action}) - \text{value of the best alternative given up}$$

Reads: the true cost of a choice is what it gives you **minus** what you gave up to get it. — **OC(action)** = the opportunity cost of doing it • **benefit(action)** = the gain the action delivers • **value of the best alternative given up** = the payoff of the *next-best* option you sacrificed by choosing this one.

- *Social media:* 1 hr/day on TikTok isn't free — you give up sport, study, work. Monetise the time with the hourly wage (avg **118 DKK** in Denmark):

YEARLY TIME-COST OF 1 HR/DAY ON TIKTOK

$$118 \text{ DKK/hr} \times 1 \text{ hr/day} \times 365 \text{ days} = \mathbf{43,070 \text{ DKK / year}}$$

Reads: price the time you *could* have been earning, added up over a year. — **118 DKK/hr** = Denmark's average hourly wage (i.e. the value of one hour of your time) • **1 hr/day** = the time you spend • **365 days** = a full year → **43,070 DKK** of wages forgone.

- *Trip to Berlin (worked example):* drive = 1,280 DKK & 14 h; fly = 1,930 DKK & 3 h. Flying frees **11 h** you could work:

OPPORTUNITY COST OF DRIVING INSTEAD OF FLYING

$$OC(\text{drive}) = (1,930 - 1,280) - (118 \times 11) = 650 - 1,298 = \mathbf{-648 \text{ DKK}} \Rightarrow \mathbf{\text{fly}}$$

Reads: the money you save by driving, minus the value of the time driving wastes. — **(1,930 - 1,280)** = the extra *ticket* cost of flying (650 DKK) • **(118 × 11)** = the value of the **11 hours** flying saves you, priced at the 118 DKK/hr wage (1,298 DKK) • a **negative** result means driving is **648 DKK worse** than flying → **fly**.

(Still a model — what if you enjoy the drive?)

Marginal thinking: decide by the value of the **next** unit — keep going while marginal benefit exceeds marginal cost; stop where they meet. That point is the **optimum**:

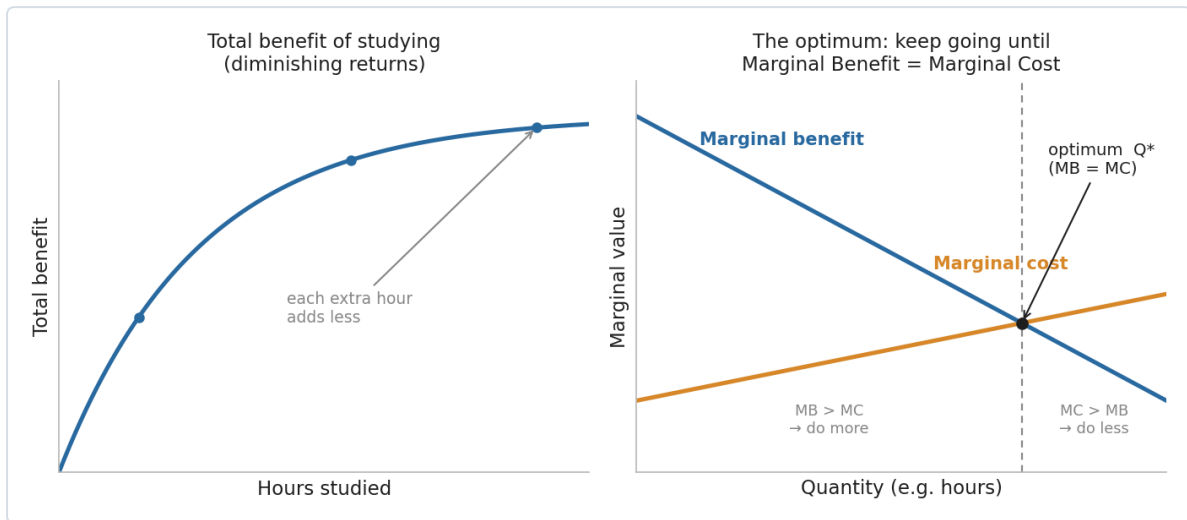
OPTIMUM CONDITION

$$\text{choose the quantity where } \mathbf{\text{Marginal Benefit} = \text{Marginal Cost}} \quad (\text{MB} = \text{MC})$$

Reads: keep going until one more unit's benefit exactly equals its cost. — **Marginal Benefit (MB)** = the *extra* benefit from **one more** unit • **Marginal Cost (MC)** = the *extra* cost of that unit • at the **optimum**

$MB = MC$ (while $MB > MC$ do more; once $MC > MB$ you've gone too far).

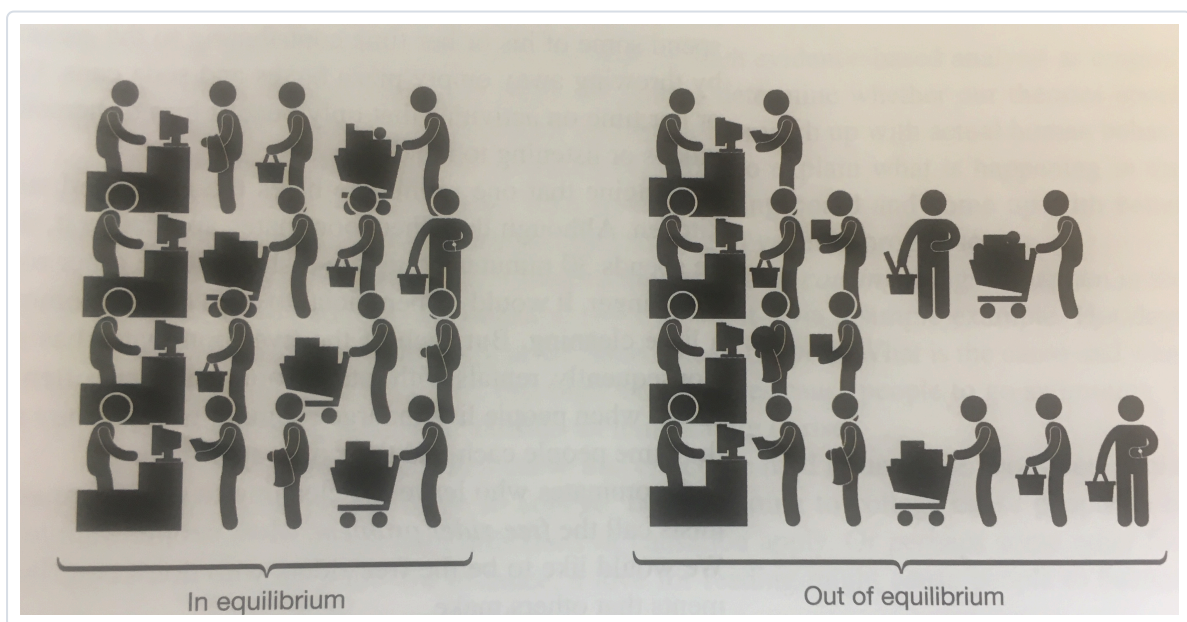
- *Night before the exam*: total benefit of studying rises then flattens; the **marginal benefit of each extra hour falls** (law of diminishing marginal returns). Optimal hours = where MB of one more hour just equals its MC.



How to read it (illustrative diagram — the graph the book draws in a later chapter). *Left*: total benefit rises but each extra hour adds less — the curve flattens (diminishing returns). *Right*: that flattening means **marginal benefit slopes down**; **marginal cost slopes up**. Left of the crossing, $MB > MC$ so one more unit is worth it (*do more*); right of it, $MC > MB$ (*do less*). The best you can do is the quantity where they meet — the optimum Q^* , where $MB = MC$.

Equilibrium

Core: an **equilibrium** is a resting point — given what everyone else is doing, **no individual can make themselves better off by changing their own behaviour**. Market prices are the classic example (later lectures): they adjust until quantity supplied = quantity demanded and no one wants to move.



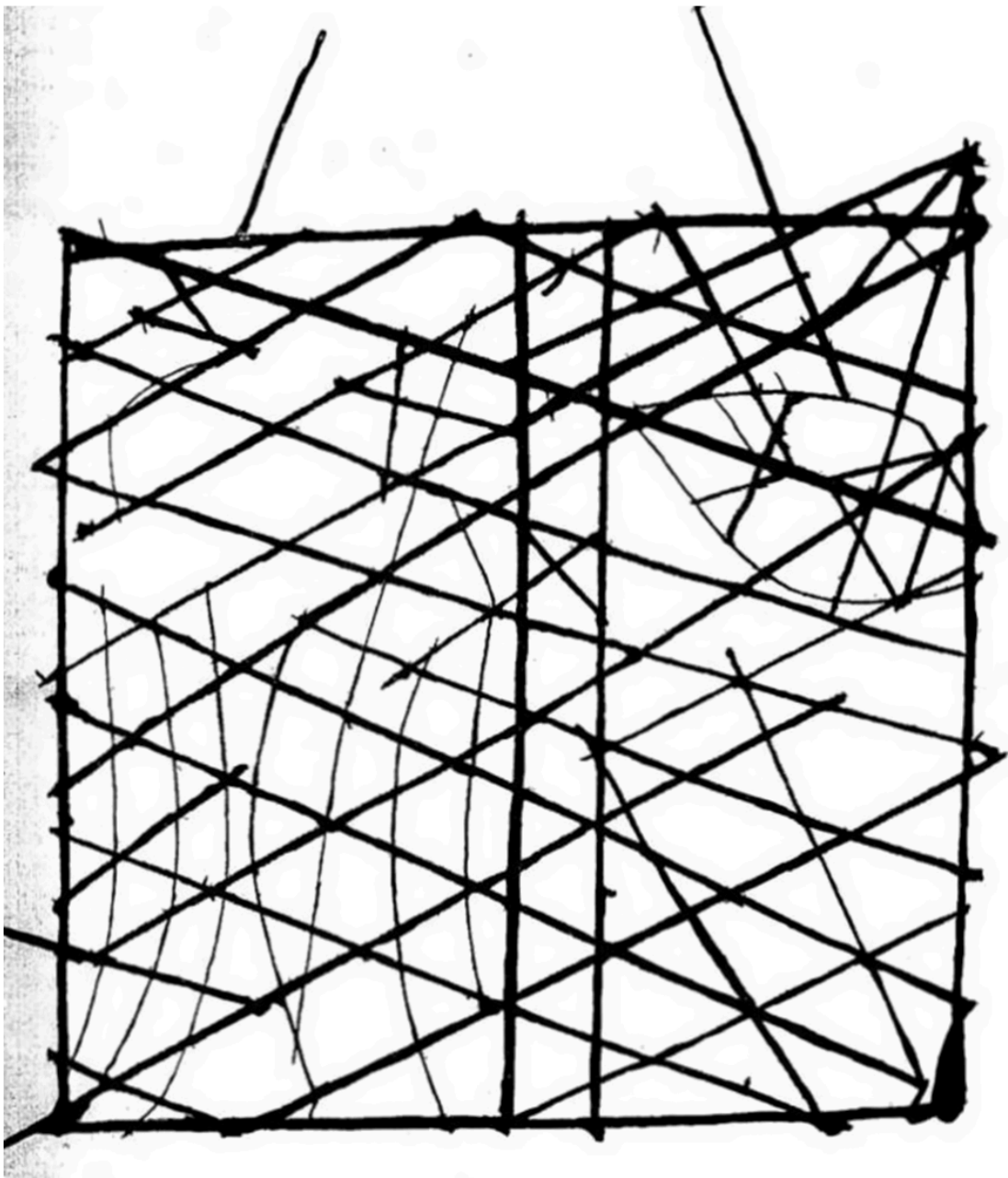
How to read it (**equilibrium** = supermarket queues, from the lecture). Each lane is a checkout; the queue length is how many people are waiting. **Out of equilibrium** (right) the lanes are uneven, so

anyone in a long line can still gain by switching to a shorter one — people keep moving. **In equilibrium** (left) all lanes are equal, so **no one can do better by changing lane on their own** — that "no incentive to move" resting point is exactly what equilibrium means (and why market prices settle where supply meets demand).

Empiricism — models, and positive vs normative

Core: economics reasons with **models** and tests them against **data** — and it deals in **positive** (testable) statements, not **normative** (value) ones.

- A **model** expresses relationships between variables and makes **simplifying assumptions** (abstracts from reality). "*All models are useful, but all models are wrong*" — the art is choosing the right model (Keynes).



How to read it (Friberg Fig. 1.1 — what a "model" is). This is a Polynesian **stick chart**: sticks map the ocean's swell patterns, shells mark islands. It's not a literal picture of the Pacific — it **strips reality down to just the features you need to navigate**. That's the book's metaphor for an economic model: deliberately simplified, "wrong" as a full description, yet **good enough to steer decisions by**.

- **Positive vs normative:**
 - **Positive** = a **testable/falsifiable** cause-effect claim — "*each additional Starbucks in a ZIP code is associated with a 0.5% rise in housing prices*" (Glaeser et al. 2018).
 - **Normative** = a **value judgment** — "*gentrification is bad because it forces the poor out.*"
 - **Economics is about positive statements**, not normative ones.
- **Correlation $\neq$ causation:** in summer people eat more ice-cream *and* drown more — ice-cream doesn't cause drowning (a hidden cause: heat). A theory is only useful if it's **empirically falsifiable**.
- **Expanding models to fit data:** "people only give to charity because they care about others' well-being" fails the data (public spending doesn't crowd out giving 1-for-1 — Andreoni 1990; people give to *inefficient* charities). So we add "**joy of giving**" (utility from the act of giving itself).

Friberg's framing — allocation mechanisms (Ch. 1's example)

Core: a scarce good (e.g. **university places**) can be allocated many ways — and each has different effects on **efficiency** and **equity**. This is the book's opening illustration of "the role of economics."

- **Non-price mechanisms:** aptitude tests/interviews · secondary-school grades · open entry + hard exams · **waiting lists** · **queues** ("camp on the sidewalk") · **lotteries** · social class/legacy.
- **Price mechanisms:** everyone pays the **same fee** (set high enough to clear demand) · **auction** places to the highest bidders.
- Real systems **combine** several (fees + tests + interviews). Micro gives the tools to analyse the **trade-offs** (who gets in, how tied to parents' income, effect on effort).

Cases & examples

Each: *the example* → *the concept it teaches*.

- **1 hr/day of TikTok** → opportunity cost ($\approx 43,070$ DKK/yr at the average wage) — nothing using scarce time is free.
- **Drive vs fly to Berlin** → opportunity cost done properly: include the *value of time*, and driving turns out to *cost* 648 DKK more.
- **Studying the night before an exam** → marginal thinking + diminishing returns; optimise where $MB = MC$.
- **Ice-cream & drowning** → **correlation $\neq$ causation**.
- **Starbucks & house prices** (Glaeser 2018) → a **positive** statement; **gentrification "is bad"** → a **normative** one.
- **Charitable giving** (Andreoni 1990) → models must be **expanded to fit data** ("joy of giving").
- **University admissions** (Friberg) → **allocation mechanisms** (price vs non-price) and their equity/efficiency trade-offs.

Key terms

Term	Meaning
Scarcity	Limited resources vs unlimited wants — the basis of economics
Optimization	Choosing the best feasible option given constraints
Equilibrium	No one gains from changing their own behaviour, given others'
Empiricism	Build models → predict → test with data
Opportunity cost	Value of the best alternative given up
Marginal thinking	Deciding by the value of one more unit
Law of diminishing marginal returns	Each extra unit adds less than the last
Model	Simplified representation of relationships between variables
Positive statement	A testable/falsifiable cause-effect claim
Normative statement	A value judgment (good/bad, should/ought)
Allocation mechanism	Rule that decides who gets a scarce good (price/lottery/queue...)

Exam pointers

- State the **three principles** (Optimization · Equilibrium · Empiricism) and define each.
- **Compute an opportunity cost** properly (include the value of time) — the Berlin example is the model answer.
- Use **marginal thinking** + $MB = MC$ to find an optimum, and name the **law of diminishing marginal returns**.
- Classify a statement as **positive vs normative**, and remember **economics = positive**; watch for **correlation $\neq$ causation**.
- Explain why economists use **models** ("all models are wrong, but useful") and give an **allocation-mechanism** trade-off (Friberg's university example).

★ Fun facts & memorable details

Sticky lines from Lecture 1.

- Keynes: "*Economics is a science of thinking in terms of models joined to the art of choosing models which are relevant to the contemporary world.*"
- The course mantra: "*All models are useful, but all models are wrong.*"
- **1 hour of TikTok a day $\approx$ 43,070 DKK a year** in forgone wages (at Denmark's $\sim$ 118 DKK average hourly wage).

- Driving to Berlin *looks* cheaper (1,280 vs 1,930 DKK) but, once you price the 11 lost hours, it actually **costs 648 DKK more** than flying.
- "During the summer people eat more ice-cream and also drown more" — the classic **correlation-≠-causation** trap (the hidden cause is heat).
- "Joy of giving" (Andreoni 1990): we don't donate only for others' welfare — we get utility from the *act* of giving, which is why public spending doesn't crowd out private giving 1-for-1.
- Learning micro is "*a first course in a new language*" — a bit hard, not immediately intuitive, needs practice.

Lecture 2: Supply and Demand (Chapter 2)

Required reading: Friberg, ch. 2 (*Supply and Demand*). • Lecturer: Luigi Butera.

Theme of the lecture (Butera): build tractable models of demand and supply, then use them to find the **market equilibrium** — the price at which "*no one would benefit from changing their own behaviour given the choices of others.*" The whole of micro is understanding that equilibrium: what sets prices and quantities, and how they move.

★ **The heart of the chapter:** a market is one picture — a **downward-sloping demand curve** and an **upward-sloping supply curve** that cross at **exactly one point, the equilibrium (p^* , q^*)**, where **quantity demanded = quantity supplied**. Three skills: ① write demand & supply as **functions**; ② set them equal to **solve for p^* and q^*** ; ③ **shift** a curve and read off the new equilibrium. *Price is set by BOTH blades of the scissors (Marshall) — never demand or supply alone.*

The red thread. For *every* question ask the one diagnostic that trips people up: **is this a *movement* ALONG a curve (only the good's own price changed) or a *SHIFT* of the whole curve (something else changed — income, other prices, costs, tastes, tech)?** Own price → slide along. Anything else → the curve moves. Get that right and the rest is algebra.

Key concepts / "modes" to use

- **Demand function** $Q_D = D(p, p_s, p_c, Y, \tau)$ → its 2-D demand curve; **inverse demand** (p as a function of q).
- **Supply function** $Q_S = S(p, p_o)$ → **supply curve**; **inverse supply**.
- **Movement along vs shift** of a curve (the core distinction); **shifters** of each curve.
- **Substitutes / complements**; **normal / inferior** goods; **Giffen** good (the rare exception).
- **Market equilibrium** $Q_D = Q_S$ → solve for p^* , q^* ; **excess demand (shortage)** / **excess supply (surplus)**.
- **Comparative statics** (shifts) & how the **slope** of the other curve sizes the effect.
- **Price ceiling / floor**; **perfect competition** assumptions; **Marshall's scissors**.

Section-by-section main points

2.1 Demand — the demand function

Core: demand answers "how much do buyers want at each price?" The **law of demand:** all else equal, **lower price → higher quantity demanded** (the curve slopes **down**).

- *In plain terms — why it slopes down:* two reasons. ① **Scarcity** — you don't have infinite money, so a lower price lets you afford more. ② **Diminishing marginal utility** — the *first* slice of pizza is worth a lot to you; the 30th almost nothing, so you'll only buy more if it's cheaper.
- **The demand function** describes quantity demanded Q_D as a function of price **and everything else**:

DEMAND FUNCTION

$$Q_D = D(p, p_s, p_c, Y, \tau)$$

Reads: how much buyers want depends on the good's own price **and** a handful of other factors. — Q_D = quantity demanded (what we're solving for) • $D(\cdot)$ = "is a function of" • p = the good's **own price** • p_s = price of a **substitute** • p_c = price of a **complement** • Y = consumers' **income** • τ = **taxes**.

- **How the 2-D curve appears** — "flush" the other variables into a constant. A demand *curve* plots Q_D against **own price only**, so you **fix** everything else at set numbers; they collapse into the intercept. The book's worked reduction — start from a specific form and plug in temperature $T=20^\circ$, substitute price $p_r=2$, income $I=5$:

COLLAPSING TO A LINEAR DEMAND CURVE

$$Q_D = 1 - p + 0.25 \cdot T + 0.75 \cdot p_r + 0.5 \cdot I = 1 - p + 5 + 1.5 + 2.5 = 10 - p$$

(intercept $a = 10$ bundles all the non-price factors)

Reads: plug fixed numbers into every non-price factor and they merge into one constant. — T = temperature ($=20$) • p_r = price of raspberries ($=2$) • I = income ($=5$); the **coefficients** (0.25, 0.75, 0.5) say how strongly each shifts demand • adding the constant pieces ($1 + 5 + 1.5 + 2.5$) gives the **intercept $a = 10$** , so only p is left as a variable → $Q_D = 10 - p$.

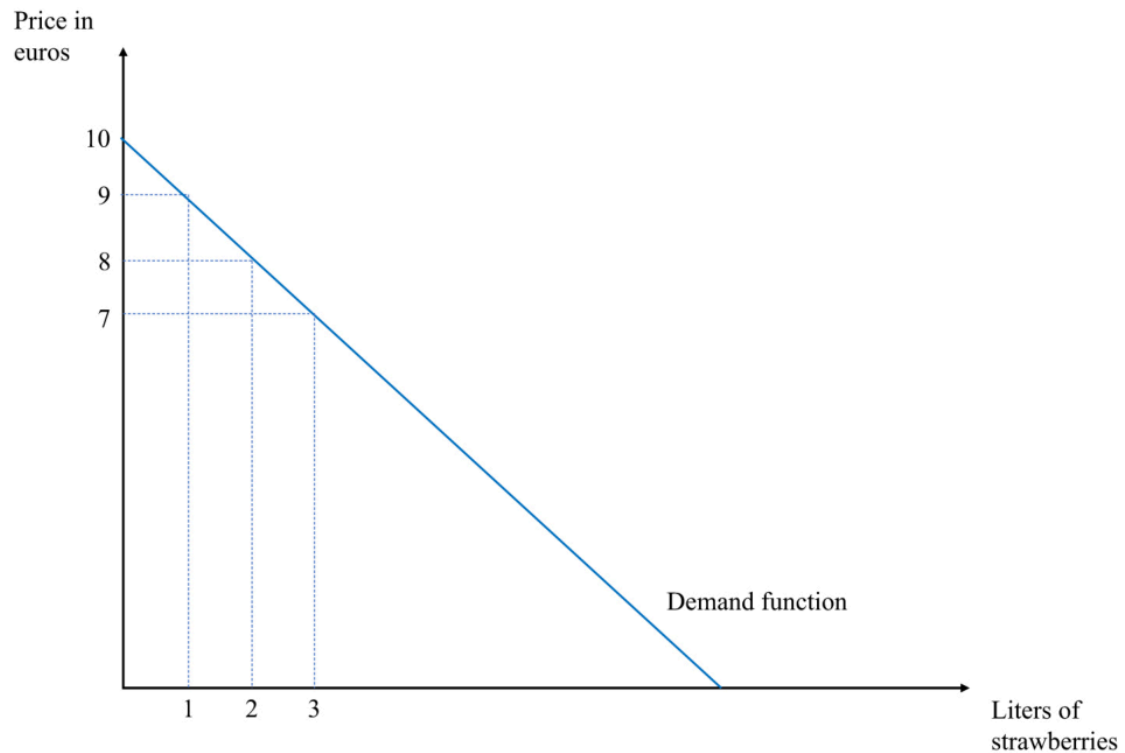
LINEAR DEMAND & ITS INVERSE (FOR GRAPHING)

$$Q_D = a - b \cdot p \rightarrow \text{e.g. } Q_D = 10 - p \text{ (} a = 10, b = 1 \text{). Solve for } p \Rightarrow \text{inverse demand: } p = 10 - Q_D \cdot \text{(deck's version: } Q_D = 20 - 2p \Rightarrow p = 10 - \frac{1}{2} \cdot Q_D \text{)}$$

Reads: a straight-line demand — quantity falls as price rises. — a = the **intercept** (quantity demanded if price were 0; it bundles income, other prices, tastes — *shift it and the whole curve moves*) • b = the **slope** (how many units demand drops per +1 in price) • p = price. The **inverse** just rearranges it to put p alone on the left, because graphs put price on the vertical axis.

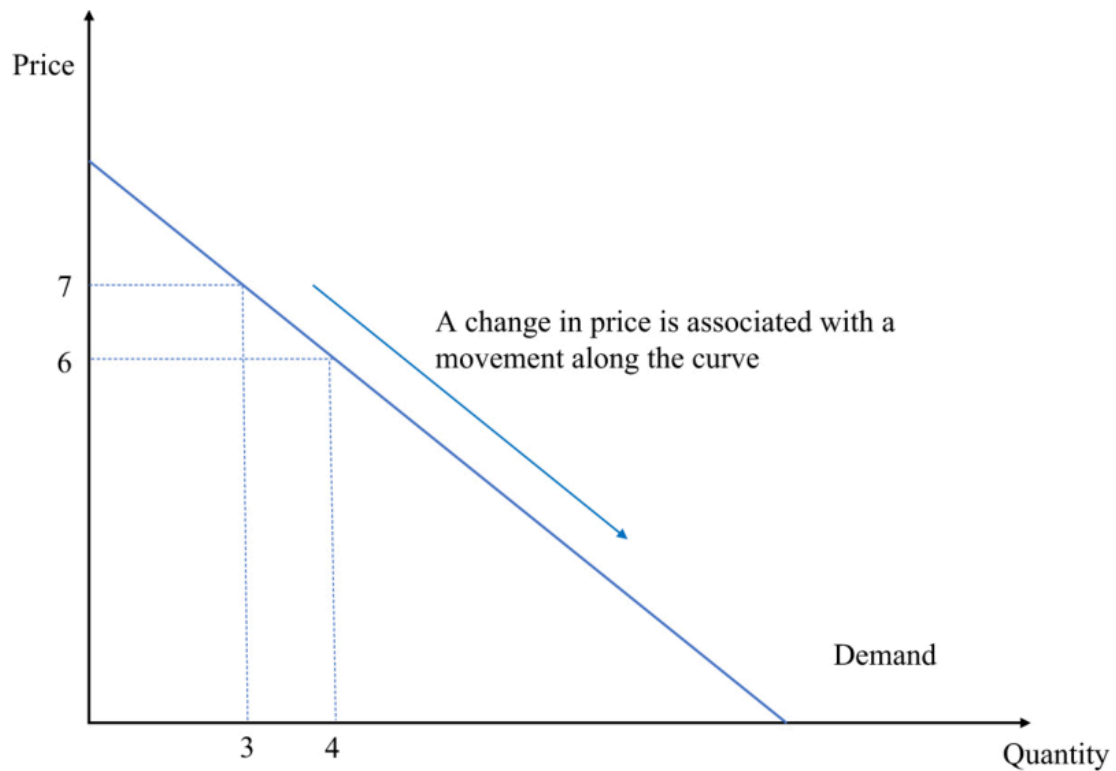
Check: $p = 7 \rightarrow Q_D = 3$; $p = 6 \rightarrow Q_D = 4$. a = the intercept (shift it and the whole curve moves); b = how many units demand falls per +1 in price.

Supply and demand



How to read it (the demand curve). By convention **price is on the vertical axis**, quantity on the **horizontal** (blame Alfred Marshall). Each point answers *both* questions: "at price p , how much is bought?" *and* "for quantity q , what's the most a buyer will pay?" Here the line hits **$p = 10$ when $q = 0$** (nobody buys above €10) and slides down to more litres as price falls.

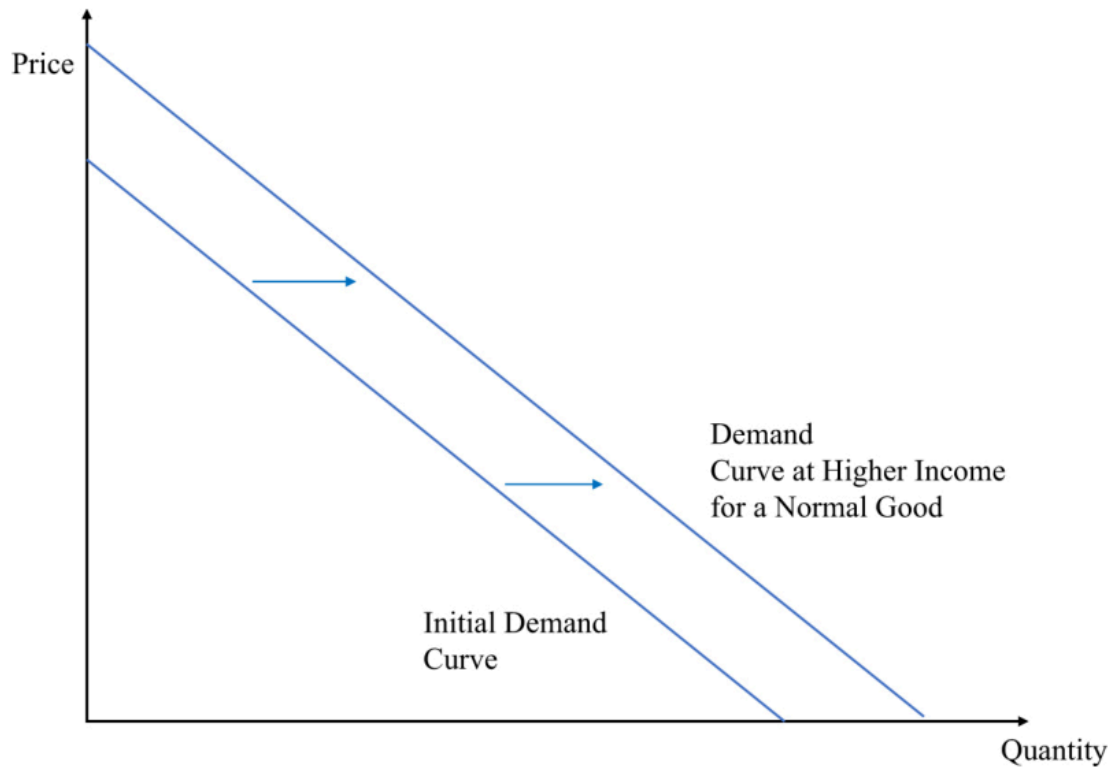
- ★ **Movement ALONG vs SHIFT — the make-or-break distinction:**
 - Own price changes → **MOVE ALONG** the existing curve (you slide to a new point).
 - Anything else changes → the whole curve **SHIFTS** (a *new* curve).



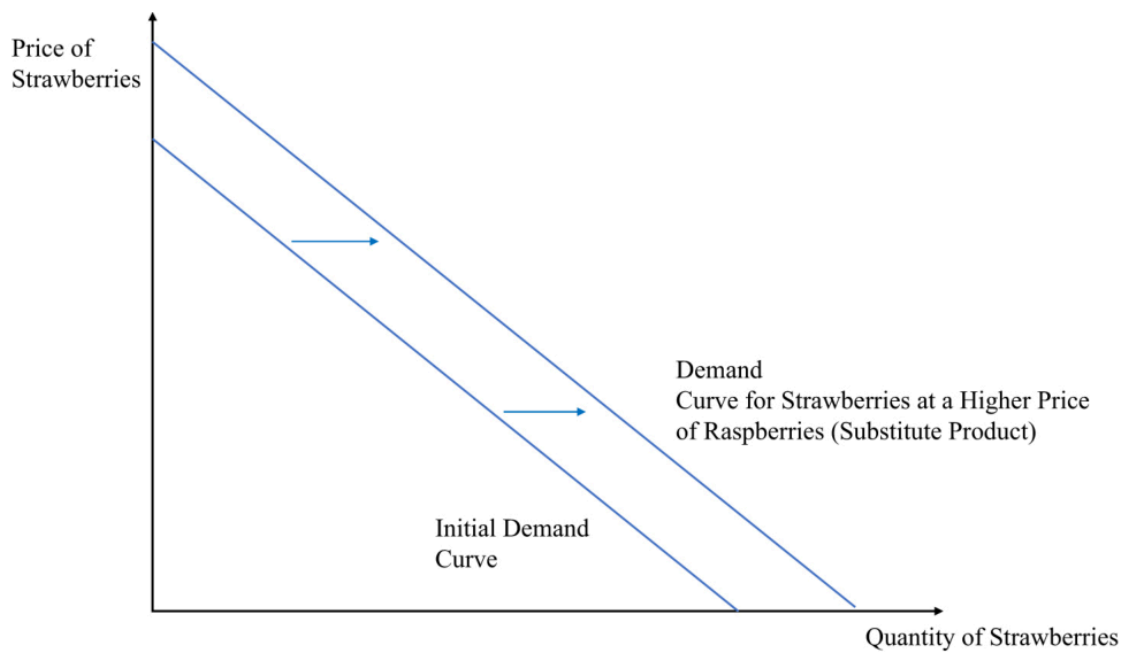
How to read it (movement along). Only the **own price** moved (7→6), so you **stay on the same curve** and just slide to a new point (q 3→4). No new curve is drawn. Contrast the shifts below, where the curve itself jumps.

- **Demand shifters** (each moves the *whole* curve = changes the intercept a):

Shifter	Direction	Why
Income ↑ (normal good)	shift out (right)	richer → buy more at every price
Income ↑ (inferior good, e.g. instant noodles)	shift in (left)	richer → buy <i>less</i> of it
Price of a substitute ↑ (raspberries ↑ → strawberry demand)	shift out	switch toward the now-relatively-cheaper good
Price of a complement ↑ (cereal ↑ → milk demand)	shift in	the pair is consumed together
Tastes / expectations (heatwave → ice-cream; COVID → toilet paper)	out or in	common-sense direction



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How to read them (shifts). The whole line moves **right (out)** = more demanded at *every* price. Fig 2.4: higher **income** (normal good). Fig 2.6: a **substitute** (raspberries) got dearer, so buyers swing to strawberries. Note you are *not* sliding along the old curve — a brand-new curve sits to the right.

- **Substitutes vs complements (definitions):** **substitutes** — demand for good 1 **rises** when good 2's price rises (raspberries ↔ strawberries). **Complements** — demand for good 1 **falls** when good 2's price rises (e.g., strawberries and jam).

price rises (milk $\leftrightarrow$ cereal).

- **From individual to aggregate demand (deck).** Market demand = **horizontal sum** of every buyer's demand — *add the quantities at each price*:

AGGREGATE (MARKET) DEMAND

$$Q(p) = q_1 + q_2 + \dots = D_1(p) + D_2(p) + \dots \text{ (add quantities across buyers at the same price)}$$

Reads: market demand = every individual's demand added up **at each price**. — $Q(p)$ = total quantity the whole market wants at price p • $q_1, q_2, \dots$ = the quantities buyer 1, buyer 2, ... each want at *that same* price • $D_1(p), D_2(p)$ = each buyer's own demand function. (It's a **horizontal** sum — fix the price, add the quantities.)

Deck example: Mike's inverse demand $p = 100 - 2q_M$, Linda's $p = 100 - \frac{1}{2}q_L \rightarrow$ invert each to $q(p)$, then add: at each price, total $q = q_M + q_L$. (Watch for a **kink** where one buyer drops out at high prices.)

- **The T-shirt trap (don't confuse products):** seeing pricey Gucci shirts outsell cheap generics is **not** an upward-sloping demand curve — those are **different goods**. A demand curve compares a good to **itself** (Gucci cutting *its own* price sells *more*). Real upward-sloping demand = the very rare **Giffen good** (Irish-famine potatoes; poor Chinese rice households — Jensen & Miller 2008).

2.2 Supply — the supply function

Core: supply answers "how much do firms bring to market at each price?" Firms are **price-takers** (they choose quantity, not the price). The curve usually slopes **up** — higher price $\rightarrow$ more supplied.

- *In plain terms — why upward:* a higher price (i) makes it worth producing more even as costs rise, and (ii) draws in higher-cost producers who couldn't profit at the low price.
- **The supply function** (quantity supplied depends on price and input/other factors):

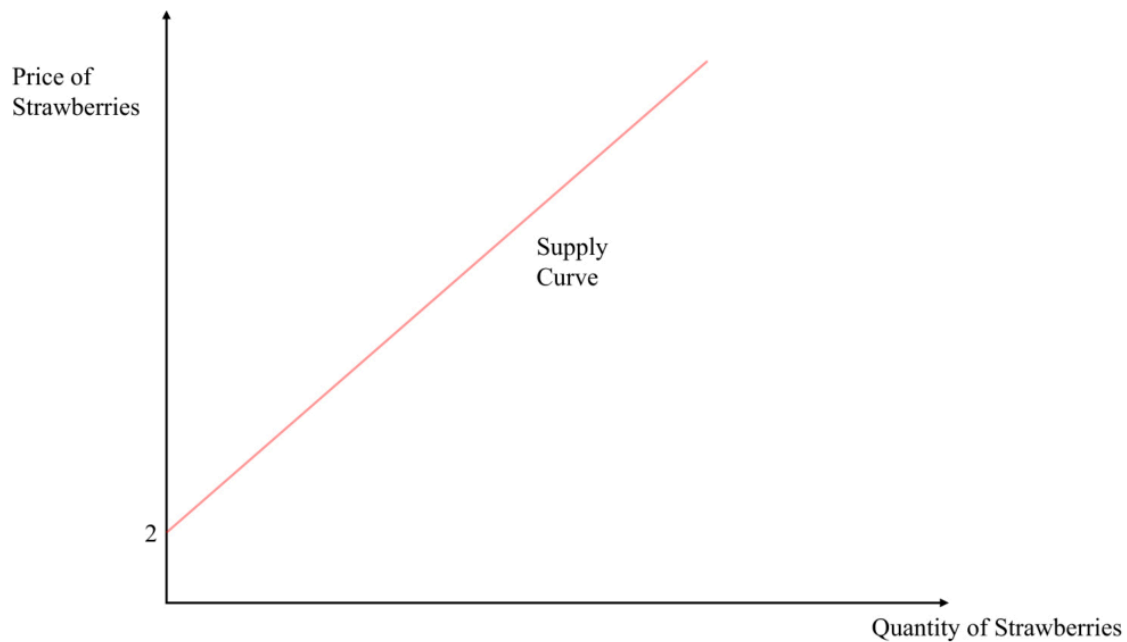
SUPPLY FUNCTION & ITS INVERSE

$$Q_S = S(p, p_o) \cdot \text{linear example: } Q_S = -2 + p \Rightarrow \text{inverse supply } p = 2 + Q_S$$

Reads: how much firms bring to market depends on the price they get and their costs. — Q_S = quantity supplied • $S(\cdot)$ = "is a function of" • p = the good's **price** • p_o = price of **inputs** (wages, materials — higher costs shift supply in). In the example the "-2" is the price-intercept read backwards: **price must exceed 2 before any is supplied** (at $p = 2, Q_S = 0$).

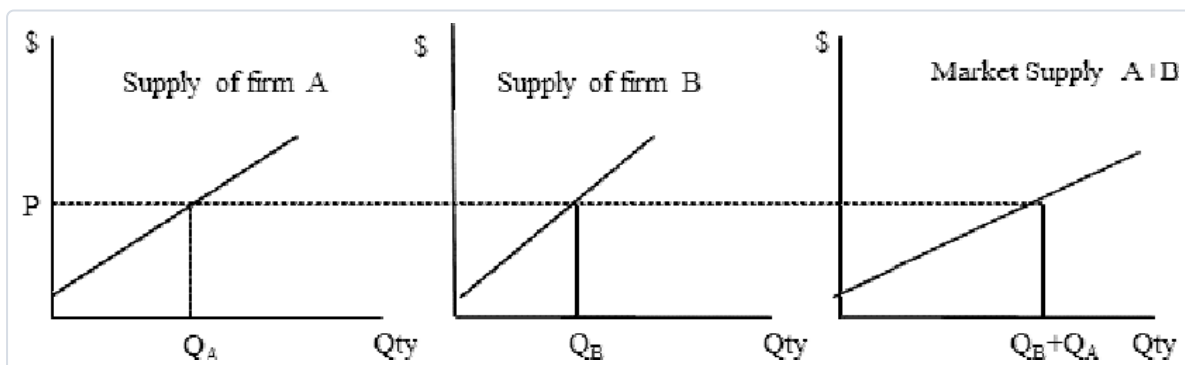
- **Why the "-2" (a negative intercept that isn't weird):** read it in **inverse** form — **price must clear €2 before any is supplied** (at $p = 2, Q_S = 0$). *Check:* $p = 3 \rightarrow Q_S = 1$; $p = 4 \rightarrow Q_S = 2$.

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How to read it (the supply curve). Start at the **€2 price-intercept** (below it, zero supply) and climb: each extra euro of price brings one more litre to market. Like demand, an **own-price** change = **move along**; anything else = **shift**.

- **Supply shifters (move the whole curve):** input/cost $\uparrow$ (wages, transport) $\rightarrow$ shift **in** (need a higher price for the same quantity); **cheaper inputs / new tech / more sellers / market opens to foreign suppliers** $\rightarrow$ shift **out**; **disruptions** (strikes, extreme weather, mine closures) $\rightarrow$ shift **in**.
- **From individual to aggregate supply (deck):** market supply = **horizontal sum** of each firm's supply — add quantities across firms at each price.



How to read it (aggregate supply). At any price **P**, read each firm's quantity off its own curve (Q_A , Q_B) and **add them across** — the market curve is the sum, so it's **flatter** (more responsive) than any single firm's.

★ 2.3 Market equilibrium — solving for p^* and q^*

Core: equilibrium = the **one price where quantity demanded = quantity supplied** ($Q_D = Q_S$), the **market-clearing price** — no shortage, no glut, nobody wants to move. **This is the calculation to master.**

- The recipe (memorise these three steps):

STEP 1 — SET DEMAND = SUPPLY

$$Q_D = Q_S \Rightarrow 10 - p = -2 + p$$

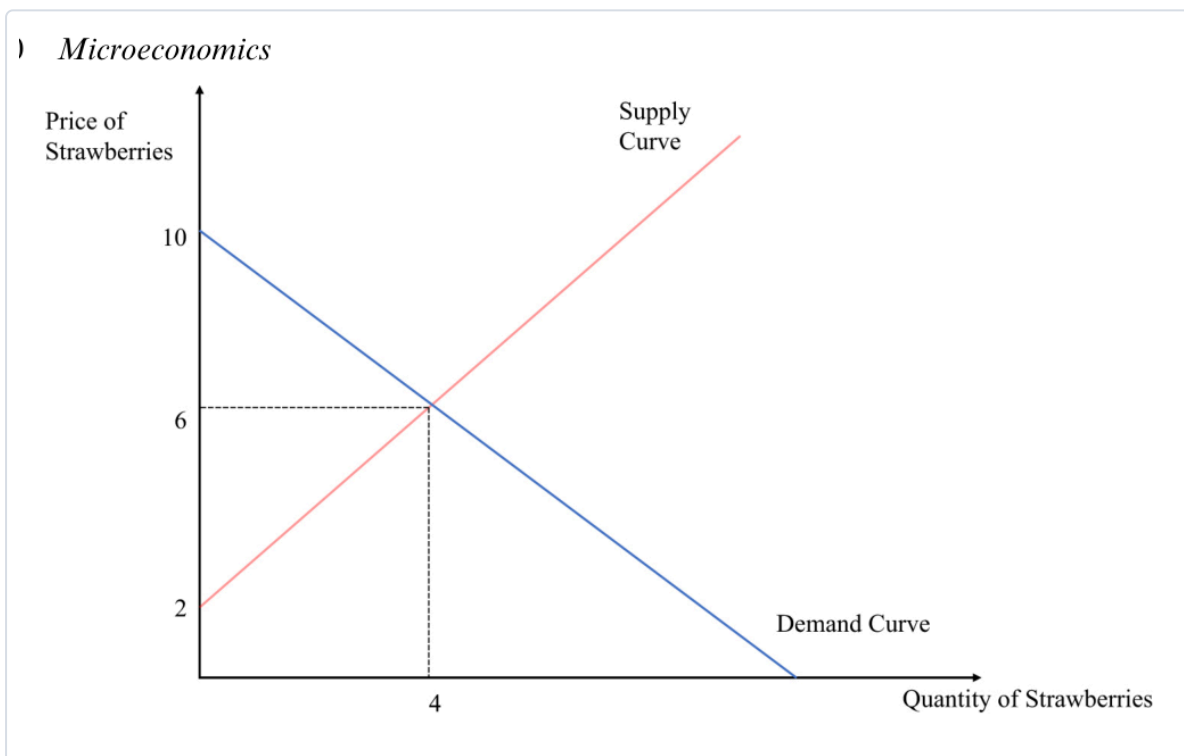
STEP 2 — SOLVE FOR THE EQUILIBRIUM PRICE P^*

$$10 + 2 = p + p \Rightarrow 12 = 2p \Rightarrow p^* = 6$$

STEP 3 — PUT P^* BACK INTO EITHER CURVE FOR Q^*

$$Q^* = 10 - 6 = 4 \text{ (check: } Q_S = -2 + 6 = 4 \checkmark) \Rightarrow \text{equilibrium } (p^*, q^*) = (6, 4)$$

Reads: find the price where the two plans agree, then the quantity that trades there. — Q_D = quantity demanded • Q_S = quantity supplied • setting $Q_D = Q_S$ solves for p^* = the market-clearing **price**; substituting p^* back into either curve gives q^* = the **quantity** traded. The star (*) just marks "equilibrium value." (Both curves must give the *same* q^* — that's the $\checkmark$ check.)



How to read it (equilibrium). The **crossing point** is the only place the two plans agree — buyers want exactly what sellers offer (4 litres at €6). Everywhere else, one side is frustrated and price gets pushed toward the cross.

- **Off-equilibrium: excess demand & excess supply** (this is also *exactly* how you handle price controls below):

EXCESS DEMAND (SHORTAGE) – PRICE BELOW P^*

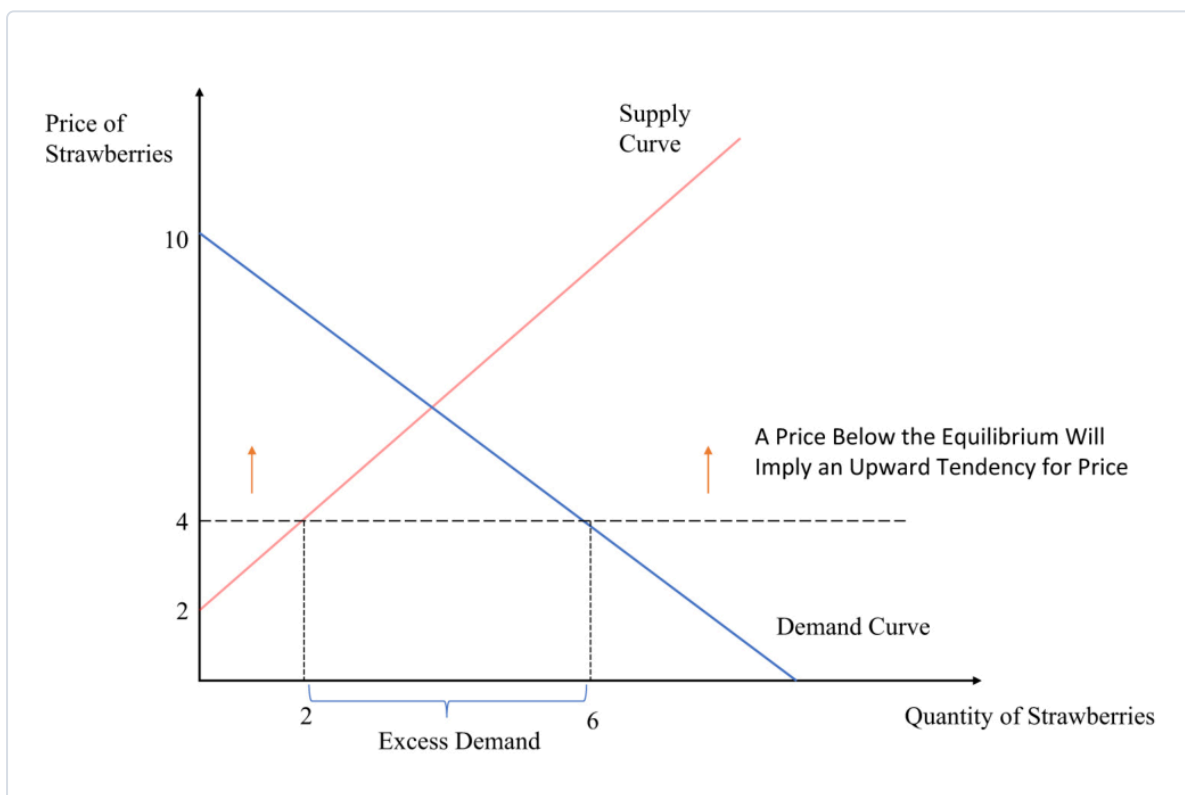
at $p = 4$: $Q_D = 10 - 4 = 6$, $Q_S = -2 + 4 = 2 \Rightarrow \text{shortage} = Q_D - Q_S = 4 \rightarrow$
upward pressure on price

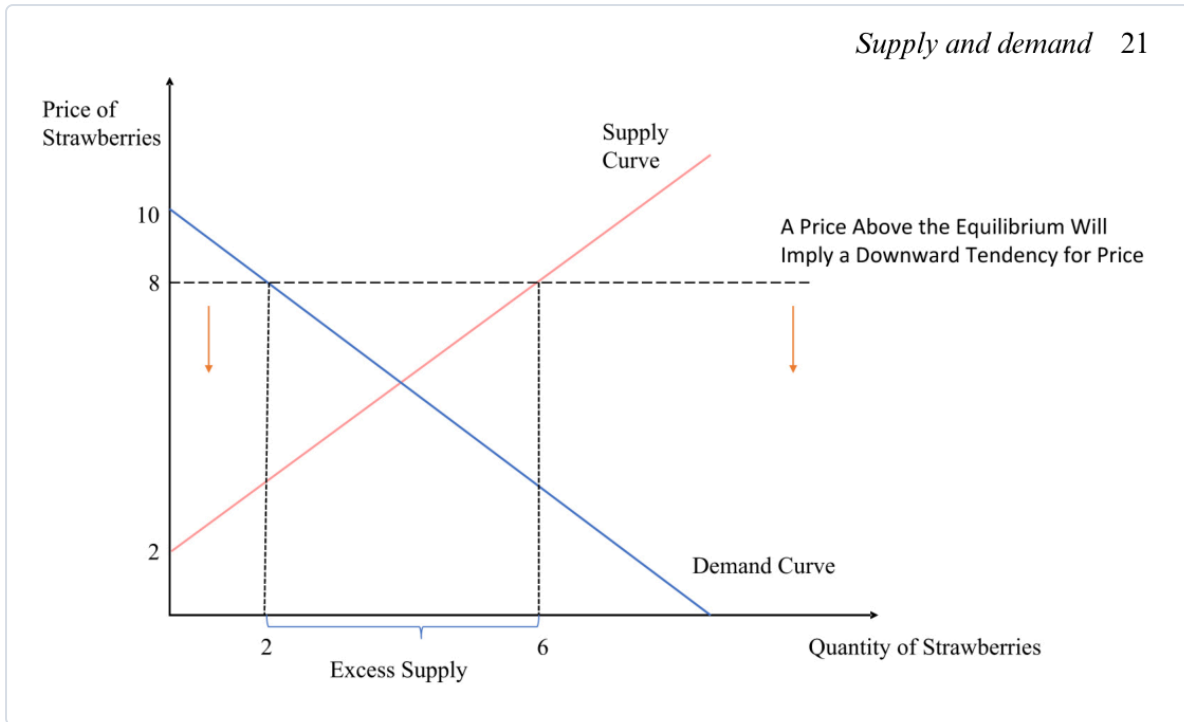
Reads: below equilibrium, buyers want more than sellers will supply. — plug the low price ($p = 4$) into **both** curves $\rightarrow Q_D = 6$ wanted, $Q_S = 2$ offered; the gap $Q_D - Q_S = 4$ is the **shortage**, which pushes the price **up** toward p^* .

EXCESS SUPPLY (SURPLUS) – PRICE ABOVE P^*

at $p = 8$: $Q_D = 10 - 8 = 2$, $Q_S = -2 + 8 = 6 \Rightarrow \text{surplus} = Q_S - Q_D = 4 \rightarrow$
downward pressure on price

Reads: above equilibrium, sellers offer more than buyers want. — plug the high price ($p = 8$) into **both** curves $\rightarrow Q_S = 6$ offered, $Q_D = 2$ wanted; the gap $Q_S - Q_D = 4$ is the **surplus**, which pushes the price **down** toward p^* .





How to read them. Pick the off-equilibrium price, drop a vertical line, and read **both** curves: the **horizontal gap** between them is the shortage (demand side wider) or surplus (supply side wider). The gap is what pushes price back toward p^* .

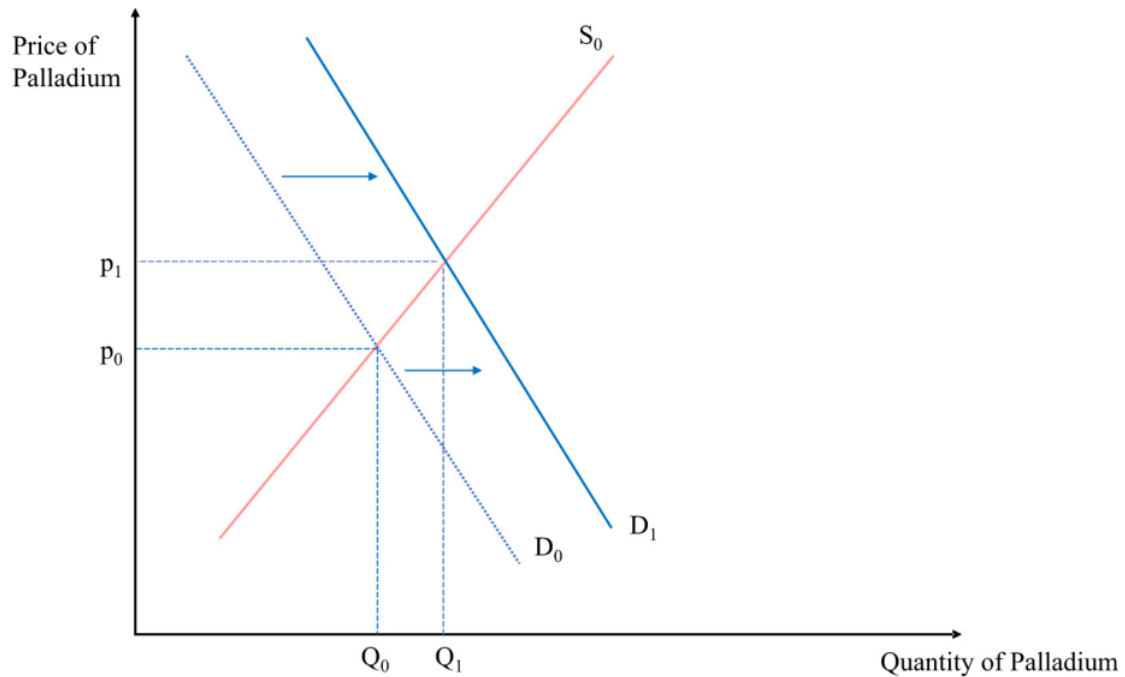
- **Marshall's scissors:** asking whether *demand or supply* sets the price is like asking *which blade of a scissors cuts* — **both do**. This dissolves the **diamond–water paradox**: water is cheap not because it's little wanted but because it's **abundant in supply**; diamonds are dear because they're **scarce**.
- (Who actually "sets" the price if everyone's a price-taker? The theoretical **Walrasian auctioneer** — and **Vernon Smith's 1962 experiments** showed real people reach the equilibrium price fast.)

2.4 Shifts in the equilibrium — comparative statics

Core: to analyse any shock, **don't jump to price/quantity**. First figure out **which curve moves and which way**; the effect on p^* and q^* follows.

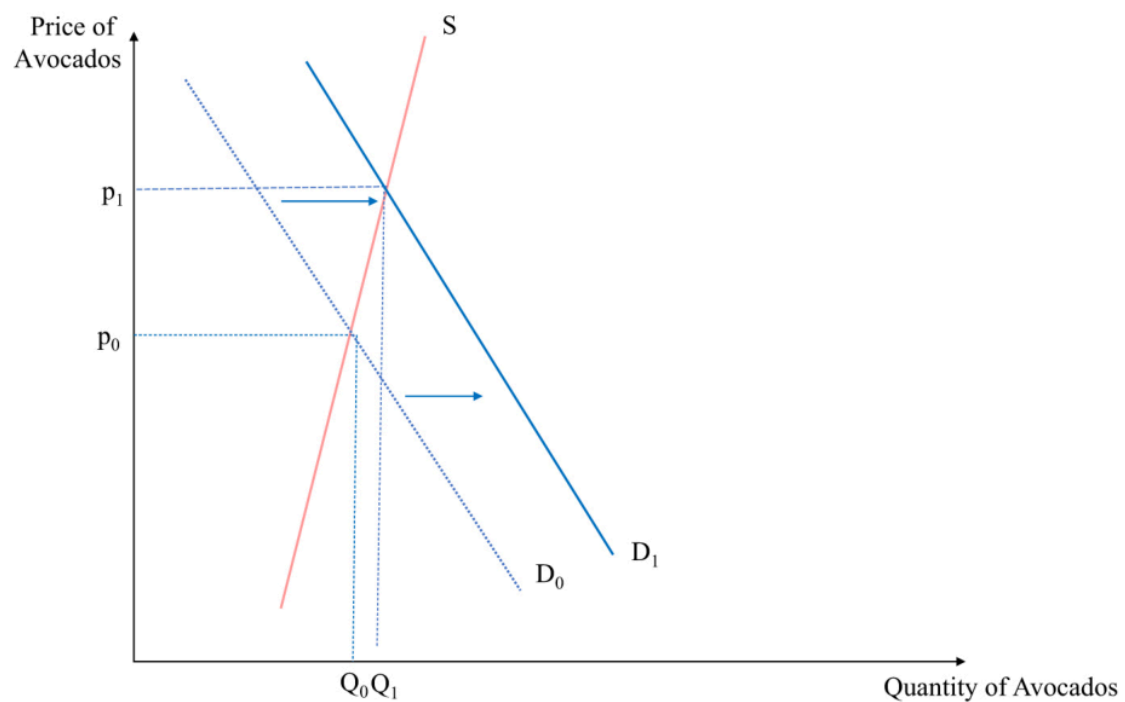
- **The method (Butera/Marshall — one change at a time, *ceteris paribus*):** ① Demand or supply? (falling incomes/other-good prices → demand; input costs/disruptions → supply). ② **Which direction** does it move *at a given price*? ③ *Then* read the new crossing. ④ **Reminder:** a *shift* is not a movement along — so a demand shift giving **both** higher price **and** higher quantity does **not** break the law of demand.

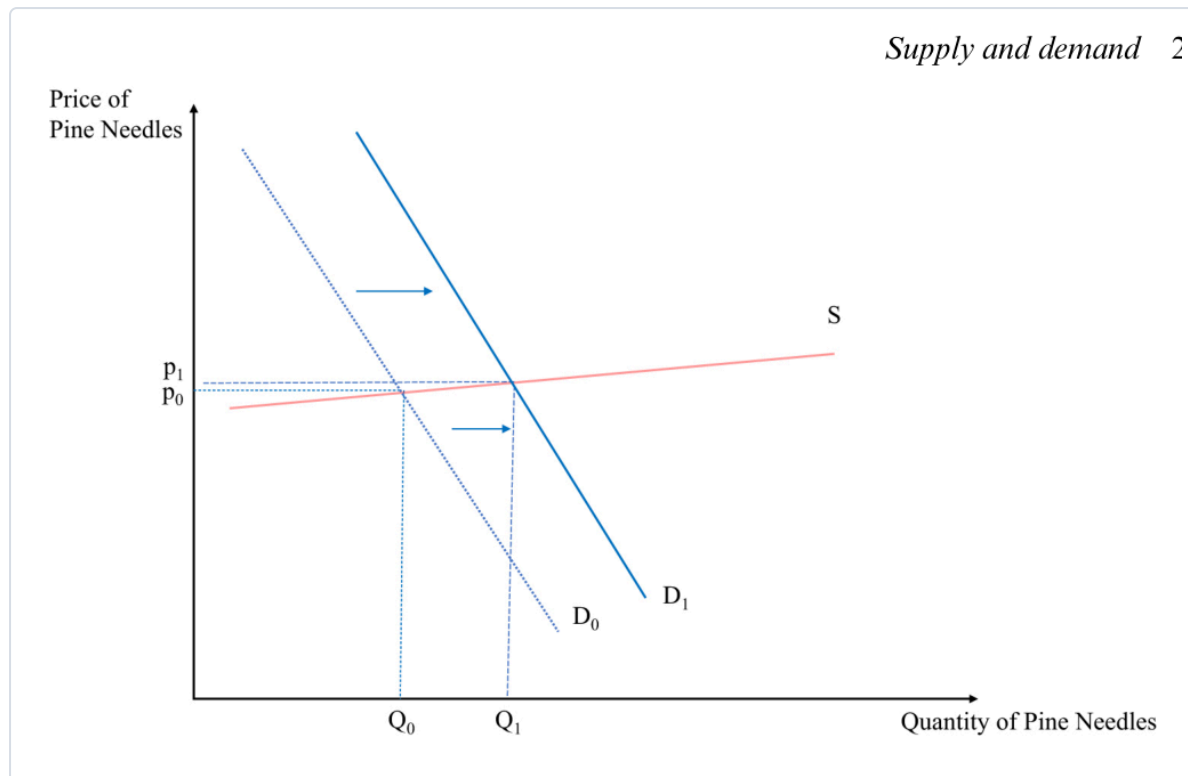
Supply and demand



How to read it (a demand shock). Demand shifts **out** ($D_0 \rightarrow D_1$); supply is unchanged, so producers **move along** supply to the new cross — **both p and q rise**. (Add a simultaneous supply **fall** and price rises *unambiguously*, but the **quantity effect becomes ambiguous** — it depends which shift is bigger.)

- ★ **Slope decides the *size* of the effect.** A demand shock's impact depends on the **slope of the supply curve**:





How to read them. Same outward demand shift, opposite results. **Steep supply** (avocados — trees take years, stock is fixed short-run): price jumps, quantity barely moves. **Flat supply** (pine needles — anyone can gather more): quantity jumps, price barely moves. *The flatter the other curve, the more a shock shows up as quantity; the steeper, the more it shows up as price.*

Price ceilings & floors — when the government fixes the price (deck)

Core: a **binding price control** stops the market clearing, and you find the resulting gap with the **exact excess-demand/supply calculation above** — just set p to the controlled price.

- **Price ceiling** = a legal **maximum** (e.g. **rent control**). To bind it must be **below p^*** , so $Q_D > Q_S$ → a **shortage** (queues, waiting lists, black-market resale). *Compute:* plug the capped p into both curves; shortage = $Q_D - Q_S$.
- **Price floor** = a legal **minimum** (e.g. the **minimum wage** in the labour market). To bind it must be **above p^*** , so $Q_S > Q_D$ → a **surplus** (unsold goods; in the labour market, **unemployment** — more people want to work than firms will hire). *Compute:* surplus = $Q_S - Q_D$.
- **The trade-offs (positive, not normative): ceilings** — **pro:** keep essentials affordable for the less-well-off; **con:** shortages, quality decline, and markets "find a way around" (rent control can even be **regressive**; Venezuela's caps produced empty shelves). **floors** — **pro:** a living wage; **con:** firms substitute **capital for labour** (self-checkout at Føtex), cut hours, or make jobs temporary. "Good economics generates **positive** statements — evidence on costs and benefits — to inform the **normative** political choice."

Is the model realistic? — perfect competition (deck)

Core: the supply-demand model assumes **perfect competition** — "every model is wrong, but useful."

- **Assumptions:** ① every agent is a **price-taker**; ② goods are **homogeneous** (perfect substitutes — one farm's wheat = another's; Shell petrol = Exxon petrol); ③ **perfect information** about prices/quality; ④ **low transaction costs**; (and many buyers & sellers). **Break these** and you need refinements — **monopoly, market failures** (later chapters).

Cases & examples

Each: *the example* → *the concept it teaches*.

- **Strawberries (the running example)** → the whole toolkit: $Q_D = 10 - p$, $Q_S = -2 + p$, equilibrium (6, 4), and the €4/€8 shortage/surplus.
- **Diamond–water paradox** → price is set by **supply and demand** (Marshall's scissors), not usefulness — water's cheap because it's abundant.
- **Palladium** (tighter emission rules → catalytic-converter demand ↑; S. African mine disruptions → supply ↓) → **comparative statics** with one and two curves moving.
- **2008 world rice spike & Haiti riots** (India's export ban cut supply; panic stockpiling shifted demand out) → a real **double shift**; prices later fell as stockpiles released + the financial crisis hit demand.
- **Instant noodles** → **inferior good**; **raspberries** → **substitute**; **cereal & milk** → **complements**.
- **Giffen goods** (Irish-famine potatoes; poor Chinese rice households) → the rare **upward-sloping demand** exception.
- **COVID toilet-paper stockpiling** → an **expectations** shift of demand.
- **Avocados vs Swedish pine-needle tea** → **steep vs flat supply** sizing a demand shock.
- **Rent control / Venezuela** → **price ceiling** → **shortage**; **minimum wage** → **price floor** → **surplus** (unemployment).

Formula sheet — quick reference

Object	Formula	Note
Demand function	$Q_D = D(p, p_s, p_c, Y, \tau)$	many variables; curve fixes all but p
Linear demand / inverse	$Q_D = a - b \cdot p / p = (a - Q_D)/b$	a = intercept (shifters), b = slope
Supply function / inverse	$Q_S = -2 + p / p = 2 + Q_S$	negative intercept = min price to supply
Aggregate (market)	$Q(p) = \sum D_i(p) ; \sum S_i(p)$	horizontal sum (add quantities)
Equilibrium	set $Q_D = Q_S \rightarrow p^*$, then q^*	the market-clearing point
Excess demand (shortage)	$Q_D - Q_S$ at $p < p^*$	→ upward price pressure; price ceiling
Excess supply (surplus)	$Q_S - Q_D$ at $p > p^*$	→ downward pressure; price floor

Exam pointers

- **Solve a market from scratch:** given a linear demand and supply, **set them equal**, get p^* , back-substitute for q^* — the strawberry case ($p^*=6$, $q^*=4$) is the model answer.

- Compute a **shortage/surplus**: plug a below- or above-equilibrium price into *both* curves and take the gap ($Q_D - Q_S$ or $Q_S - Q_D$) — and know a **ceiling**→shortage, **floor**→surplus.
- Diagnose movement vs shift, and for a shock say **which curve moves, which way**, and the effect on p^* & q^* (incl. the **ambiguous-quantity** double-shift and the **slope** point).
- Explain **why demand slopes down** (scarcity + diminishing marginal utility) and **Marshall's scissors** / the diamond–water paradox.
- Name the **perfect-competition** assumptions and why the model is still "useful though wrong."

★ Fun facts & memorable details (Lecture 2)

Sticky bits from Supply & Demand.

- **Marshall's scissors**: asking whether demand or supply sets the price is like asking **which blade of the scissors** does the cutting — both do.
- The **diamond–water paradox** is only a paradox if you forget supply: water is life-or-death yet cheap **because it's abundant**.
- **Giffen goods** really exist: when their staple got dearer, poor Chinese rice households bought **more** rice, not less (Jensen & Miller 2008) — they could no longer afford anything else.
- A negative supply intercept ($Q_S = -2 + p$) isn't a mistake — it just means **price must top €2 before anyone supplies a drop**.
- **Vernon Smith** won a Nobel (2002) for showing that real people in a lab **converge on the equilibrium price** startlingly fast — the "Walrasian auctioneer" made flesh.
- Rent control's dirty secret: it can be **regressive** and spawn black-market "key money" — a "tale of good intentions" (cf. Venezuela's empty shelves).
- Minimum wage as a price floor nudges firms toward **capital over labour** — the **self-checkout at Føtex** is a supply-and-demand diagram in the wild.

Lecture 3: Elasticities & Taxes — the Supply & Demand model II (Ch. 2 recap · §6.4)

Required reading: Friberg, §6.4 Elasticities (own-price, cross-price, income; supply elasticity) — building on the **Chapter 2** supply-and-demand model (see [Lecture 2](#)). *The lecture deck also teaches taxes on goods & services — that part draws on the textbook's §7.5 (unit tax, tax incidence), covered here because the deck foregrounds it.*

Theme of the lecture (Butera — from the deck, titled "*Supply and Demand model*"). Chapter 2 taught us to **move the curves around**; this lecture adds **precision**. The *"shape"* of a demand or supply curve — its *elasticity* — decides how much a shock (or a tax) shows up as a change in price versus a change in quantity. The deck's

three goals: (1) understand what demand & supply **elasticities** mean, (2) **derive** them from demand/supply curves, and (3) use them to analyse the **effects of taxes**.

★ **The heart of the lecture: elasticity = % responsiveness** — "by what % does quantity change when price rises 1%." It's a **unit-free** number that summarises a curve's **shape**, so it's comparable across markets and time. The **shape decides how a shock splits into price vs quantity**: with **inelastic** demand a shock/tax hits **price**; with **elastic** demand it hits **quantity**. And a **tax** drives a **wedge** between the price buyers pay and sellers receive — **who bears it depends on the relative elasticities**, and (in perfect competition) **it doesn't matter who is legally taxed**.

The red thread. For any market ask: *how responsive is quantity to price (elastic or inelastic)?* — because that single question answers **how a supply shock moves the market**, **how a firm should price**, and **who really pays a tax**. Elasticity is the "shape" number that ties all three together.

Key concepts / "modes" to use

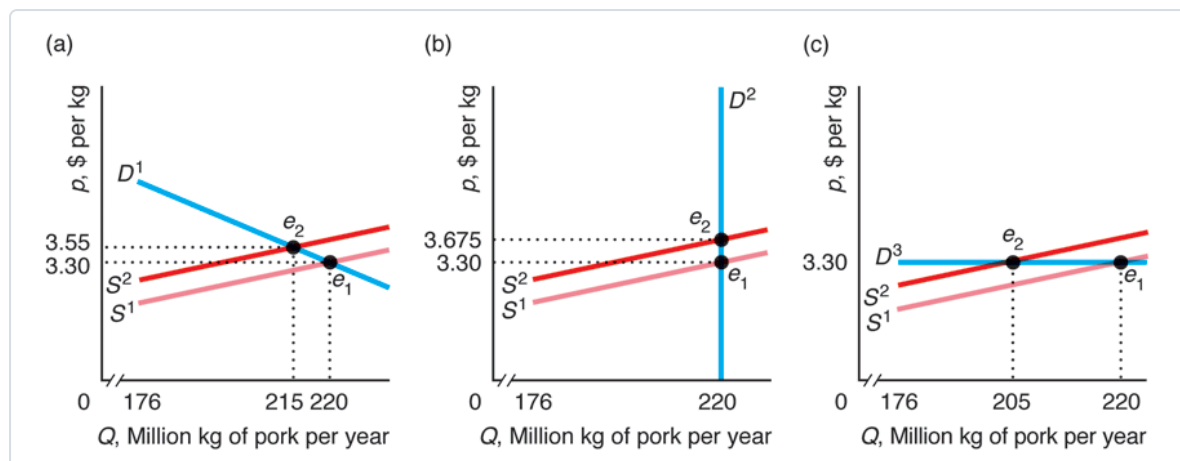
- **Elasticity** — the % change in one variable ÷ % change in another (a unit-free number).
- **Price elasticity of demand (E_D)** — % change in quantity demanded ÷ % change in own price (negative).
- **Elastic / inelastic / unit-elastic** — $|E| > 1$ / $|E| < 1$ / $|E| = 1$.
- **Point-elasticity trick** — $E_D = (dQ/dp) \cdot (p/Q)$; on a linear curve elasticity **changes along the curve** (constant-elastic curves don't).
- **Cross-price elasticity** — substitutes (+) vs complements (-).
- **Income elasticity** — normal (+) vs inferior (-); **Engel curves / Engel's law**.
- **Supply elasticity** — % change in quantity supplied ÷ % change in price (positive).
- **Tax wedge & tax incidence** — $p_b = p_s + t$; who bears the burden depends on **relative elasticities**; legal $\neq$ economic incidence; unit vs ad-valorem tax; **deadweight loss**.

Section-by-section main points

Why do we care about elasticity? (deck)

Core: the **shape** of demand/supply curves determines **how much a shift in one moves the equilibrium** — and "shape" is summarised by **elasticity**. A steep (inelastic) curve and a flat (elastic) curve respond to the *same* shock very differently.

- **The motivating questions (deck):** Do "**sin taxes**" (cigarettes, sugar) hurt the poor? Why do firms **raise prices** even when it looks crazy (Netflix, Amazon Prime)? To what extent are **taxes on producers "passed on"** to consumers? Every answer runs through elasticity.
- **The visual intuition:** how far a **supply shift** moves price vs quantity depends entirely on the **slope/elasticity of demand** — see the figure below (perfectly inelastic demand → only price moves; perfectly elastic demand → only quantity moves).



How to read it (why elasticity matters). All three panels apply the **same supply shift** $S^1 \rightarrow S^2$. The **only thing** that differs is the **demand curve's shape**:

- **(a) Ordinary downward-sloping demand:** the shock splits into **both** a higher price *and* a lower quantity.
- **(b) Perfectly inelastic demand (vertical):** quantity can't respond, so the shock lands **entirely on price** ($3.30 \rightarrow 3.675$). *This is the "necessity with no substitutes" case.*
- **(c) Perfectly elastic demand (horizontal):** price can't move, so the shock lands **entirely on quantity** ($220 \rightarrow 205$).
- **Take-away:** the **more inelastic** the demand, the more a shock (or tax) shows up as **price**; the **more elastic**, the more it shows up as **quantity**. *This one picture is the whole reason we measure elasticity.*

★ Price elasticity of demand — definition

Core: the **price elasticity of demand** measures how sensitive quantity demanded is to the good's own price — the **% change in quantity** $\div$ **% change in price**. It is a **pure (unit-free) number** and is **negative** (demand slopes down: price up $\rightarrow$ quantity down).

PRICE ELASTICITY OF DEMAND

$$E_D = (\Delta Q / Q \times 100) \div (\Delta p / p \times 100) = \% \Delta Q \div \% \Delta p$$

Reads: how responsive quantity demanded is to a price change — "a 1% rise in price p leads to an E_D % change in quantity q ." — $\Delta Q / Q \times 100$ = the **percentage change in quantity demanded** · $\Delta p / p \times 100$ = the **percentage change in price** · E_D = the ratio (a **unit-free number**, **negative** for a normal good). The $\times 100$ s cancel, so it's just **% ΔQ over % Δp** .

- **Why unit-free matters (the point of the whole concept):** raw slopes aren't comparable — "1,000 tons less rice" means something different in **Vietnam** vs **Bhutan**, and depends on the currency. A **percentage measure** is **comparable across markets, currencies and time**.
- **Why negative:** demand curves slope **down**, so ΔQ and Δp have opposite signs $\rightarrow$ the ratio is negative. (When people say a "higher" elasticity they usually mean a bigger **absolute value** $|E_D|$.)
- **Worked example:** a café raises its coffee price **10%** and sells **8% fewer** cups $\rightarrow E_D = -8\% \div +10\% = -0.8$. Since $|-0.8| < 1$, demand is **inelastic** — revenue actually **rises** (price up 10%, quantity down only 8%). (Full step-by-step computations are in the next section.)

Computing elasticity — and the point-elasticity trick

Core: plug the two percentage changes into the formula. On a linear demand curve the elasticity is *not* constant — it depends on *where* you evaluate it, via $E_D = (dQ/dp) \cdot (p/Q)$.

- **Simple example (textbook Table 6.2):** quantity falls $6 \rightarrow 4$ while price rises $1 \rightarrow 2$. Then

WORKED — SIMPLE EXAMPLE

$$E_D = ((4-6)/6) \div ((2-1)/1) = (-2/6) \div (1) = -1/3$$

Reads: a demand that is **inelastic** — $(4-6)/6 = -0.33$ = the % fall in quantity (-33%) $\cdot$ $(2-1)/1 = 1$ = the % rise in price ($+100\%$) $\cdot -1/3$ = quantity falls **only** $\frac{1}{3}$ as fast as price rises $\rightarrow$ **inelastic** demand.

- **A policy-style example (soda, Allcott et al. 2019):** own-price elasticity of sugary soda ≈ -1.37 . If price rises 5%, quantity changes by $E_D \times 5\% = -1.37 \times 5 = -6.85\%$ (a $\sim 7\%$ fall). To cut demand 15%, price must rise $-15 \div -1.37 \approx 11\%$. $\rightarrow$ this is exactly the sum a policymaker designing a **soda tax** needs.
- **The point-elasticity trick (linear demand):** rewrite the formula as

POINT ELASTICITY

$$E_D = (\Delta Q / \Delta p) \cdot (p/Q) = (dQ/dp) \cdot (p/Q)$$

Reads: elasticity = **slope of the demand function** $\times$ the **price-to-quantity ratio at your chosen point** — dQ/dp = the **slope** (how many units Q changes per €1 of price; constant for a straight line) $\cdot$ p/Q = the **price-to-quantity ratio** where you evaluate it (this is what varies along the curve).

- **Worked ($Q = 10 - 2p$):** here $dQ/dp = -2$ (always). But the **elasticity varies**:
 - at $p = 2.5$ (so $Q = 5$): $E_D = -2 \times 2.5/5 = -1$ (**unit elastic**).
 - at $p = 4$ (so $Q = 2$): $E_D = -2 \times (p/Q) = -2 \times 4/2 = -4 \rightarrow$ **elastic**. ($\triangle$ The textbook misprints this as $-2 \times 2/2 = -2$ — it accidentally puts $Q=2$ in the numerator instead of $p=4$; the correct p/Q is $4/2 = 2$, so $E_D = -4$.)
 - **The pattern: demand is more elastic at higher prices** — at $p=2.5$ it's unit-elastic (-1), at $p=4$ it's elastic (-4). A fixed unit change is a bigger *percentage* change when Q is small and p is large.
- **Constant-elasticity demand (contrast):** the form $Q = a \cdot p^\epsilon$ has the **same** elasticity ϵ at *every* price (used in empirical estimation via $\ln Q = \ln a + \epsilon \cdot \ln p + \text{error}$). So: **linear $\rightarrow$ elasticity changes along the curve; constant-elastic $\rightarrow$ it doesn't**.
 - **Worked example:** with $Q = 100 \cdot p^{-1.2}$, the elasticity is -1.2 whether you evaluate at $p = €1$ ($Q = 100$), $p = €5$ ($Q \approx 15$), or $p = €50$ — it never changes. (Contrast the linear $Q = 10 - 2p$ above, where it swung from -1 to -4 .)

Elastic, inelastic, unit-elastic — and why it matters

Core: we name the ranges of elasticity, and each has a sharp economic meaning for how supply shocks split into price vs quantity.

Name	Level (of E_D)	Meaning	Curve looks...
Inelastic	$-1 < E_D < 0$ ($ E < 1$)	%ΔQ smaller than %Δp	relatively steep
Unit elastic	$E_D = -1$	%ΔQ equals %Δp	—
Elastic	$E_D < -1$ ($ E > 1$)	%ΔQ bigger than %Δp	relatively flat

- **Why it matters (the punchline):** with **inelastic** demand, **supply shocks translate into big price changes** (little quantity response) — critical for e.g. **cocoa** growers in Ghana/Côte d'Ivoire (world cocoa elasticity ≈ -0.19 to -0.96 ; policy point estimate -0.34). With **elastic** demand, shocks translate into **quantity** changes.
- **What makes demand elastic — substitutes.** The single biggest driver is **availability of close substitutes**: pharmaceuticals with few substitutes are **inelastic** (people pay whatever for a needed medicine); goods with many substitutes are **elastic**.
- **Aggregation level matters:** a *specific brand* is more **elastic** than the *category* (more substitutes for one brand). US beer: **product-level** median elasticity ≈ -4.74 , but beer **overall** ≈ -0.60 .

Cross-price elasticity — substitutes vs complements

Core: the **cross-price elasticity** measures how demand for good **A** responds to a change in the price of a **different** good **B**. Its **sign** tells you the *relationship* between the two goods; its **size** tells you *how strong* that relationship is.

CROSS-PRICE ELASTICITY

$$E_{cross} = \% \Delta Q_A \div \% \Delta p_B \rightarrow > 0 = \text{substitutes} \cdot < 0 = \text{complements} \cdot \approx 0 = \text{unrelated}$$

Reads: how the quantity of **A** responds to a price change in **B** — %ΔQ_A = percentage change in demand for good A · %Δp_B = percentage change in the price of good B · E_cross = the ratio (can be + or –, unlike own-price which is always –).

- **The sign = the relationship:**
 - **Positive → substitutes.** B gets dearer → buyers switch to A → A's demand **rises** (butter ↑ → margarine demand ↑; Heineken ↑ → Carlsberg ↑).
 - **Negative → complements.** B gets dearer → buyers use less B *and* less of its partner A → A's demand **falls** (petrol ↑ → big-car demand ↓; smartphones ↑ → phone-case demand ↓).
 - $\approx 0 \rightarrow$ **unrelated** goods (the price of bread barely touches demand for haircuts).
- **The size = the strength:** the larger the **absolute value** $|E_{cross}|$, the **closer** the substitutes (or the stronger the complementarity). Two brands of the *same* soda have a **big** positive cross-price elasticity; soda and bottled water a **small** one; soda and umbrellas ≈ 0 .
- **Worked example:** Danish spirits get **10% dearer** → Swedish demand for spirits changes by $E_{cross} \times 10\% = +0.3 \times 10 = +3\%$ (Swedes buy **more at home** instead of driving to Denmark) → **positive** → **substitutes** (Asplund 2007). *Flip it: if smartphones get 10% dearer and phone-case demand falls 6%,* $E_{cross} = -6 \div 10 = -0.6 \rightarrow$ **negative** → **complements**.

- ★ **Its killer application — defining markets & judging mergers.** This is *the* reason cross-price elasticity matters in practice. Competition authorities use it to decide **whether two products are in the same market**: a **high positive** cross-price elasticity means A and B are **close substitutes**, so a firm selling both has real market power (a merger between them removes a competitor). This is the logic of the **SSNIP / "hypothetical monopolist" test** — *would a Small but Significant Non-transitory Increase in Price (say 5%) be profitable, or would enough buyers flee to substitutes to make it a loss?* If they flee, the substitute is in the same market.
- Asymmetry (watch out):** cross-price elasticity is **not symmetric** — A's response to B's price $\neq$ B's response to A's price. A cheap **complement** (phone case) reacts strongly to the price of the expensive main good (phone), but the phone barely reacts to the case's price.
- Real numbers (Asplund et al. 2007):** Swedish **spirits +0.3, beer +0.5, wine +0.2** with respect to their **Danish** prices — all **positive**, i.e. all **substitutes** for cross-border shopping (beer the closest). (*Deck aside: cigarettes & alcohol ≈ -1 — strong complements, Krauss et al. 2014.*)

Income elasticity — normal vs inferior goods, Engel curves

Core: the **income elasticity** measures how demand responds to **income** — its **sign** distinguishes **normal** from **inferior** goods.

INCOME ELASTICITY

$$E_{income} = \% \Delta Q \div \% \Delta I \rightarrow > 0 = \text{normal good} \cdot < 0 = \text{inferior good}$$

Reads: how quantity demanded responds to an income change — $\% \Delta Q$ = percentage change in quantity · $\% \Delta I$ = percentage change in income · **sign: positive $\rightarrow$ normal** (demand rises with income), **negative $\rightarrow$ inferior** (demand falls as income rises — e.g. instant noodles, store-brand goods).

- Worked example:** a household's **income rises 10%** and its spirits purchases rise **14%** $\rightarrow E_{income} = +14 \div +10 = +1.4 \rightarrow$ **normal and income-elastic** (a "luxury," > 1) (Asplund 2007). *Flip it: income +10% but instant-noodle demand -5% $\rightarrow E_{income} = -5 \div 10 = -0.5 \rightarrow$ inferior.*
- Nuance:** a good can be **normal at low incomes and inferior at higher ones** (hostels for a student: more nights as income rises, then switch to hotels). Tastes vary by person.
- Uses:** long-term **projections** (healthcare, appliances in low-income countries) and predicting the **business cycle** (pharma is income-insensitive $\rightarrow$ "defensive stock").
- Engel curves / Engel's law (Ernst Engel, 1800s Prussian statistician):** plot demand (vertical) against income (horizontal) — **positive slope for normal goods, negative for inferior**. **Engel's law:** the **share of income spent on food falls** as income rises $\rightarrow$ the income elasticity of food is **less than 1**.

Supply elasticity — same principle

Core: the **price elasticity of supply** measures how sensitive quantity *supplied* is to price — same formula, but **positive** (supply slopes up).

PRICE ELASTICITY OF SUPPLY

$$E_S = \% \Delta Q_S \div \% \Delta p \quad (\text{positive})$$

Reads: how responsive quantity supplied is to a price change — $\% \Delta Q_S$ = percentage change in quantity supplied · $\% \Delta p$ = percentage change in price · E_S = the ratio (**positive**: higher price → more supplied). A **steep** supply curve is **inelastic** (hard to expand output quickly); a **flat** one is **elastic**.

- **Worked example:** the market price of new flats rises **20%**, but the quantity of new flats supplied rises only **4%** this year → $E_S = +4 \div +20 = +0.2$ → **inelastic** supply (you can't build quickly). Give it several years and builders respond more → supply becomes **more elastic**.

Elasticity over time — short run vs long run

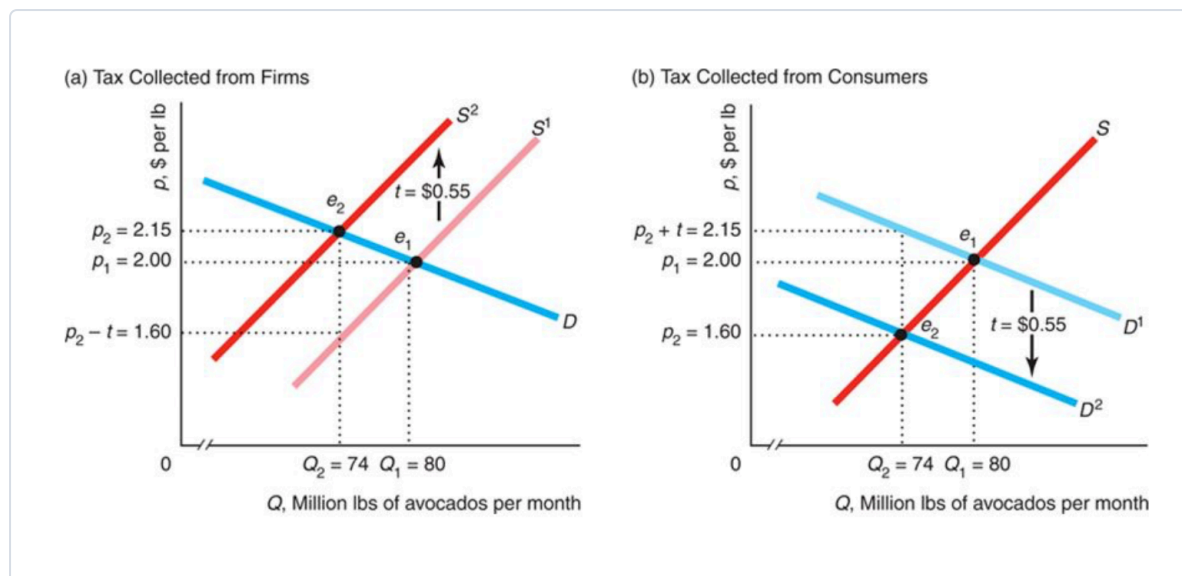
Core: demand (and supply) are usually **more elastic in the long run** than the short run, because people have **more time to adjust / find substitutes**.

- **Electricity (Labandeira et al. 2017):** ≈ -0.24 short-run vs -0.6 long-run — over time consumers buy more efficient equipment, change habits.
- **The deck's cases:** a drop in **gas** prices does little in the short run but more in the long run; likewise **computers**. *Exam framing:* "in the short run demand is inelastic (people are locked in); given time they substitute, so long-run demand is more elastic."

★ Taxes on goods & services — the wedge, incidence & equivalence (deck; textbook §7.5)

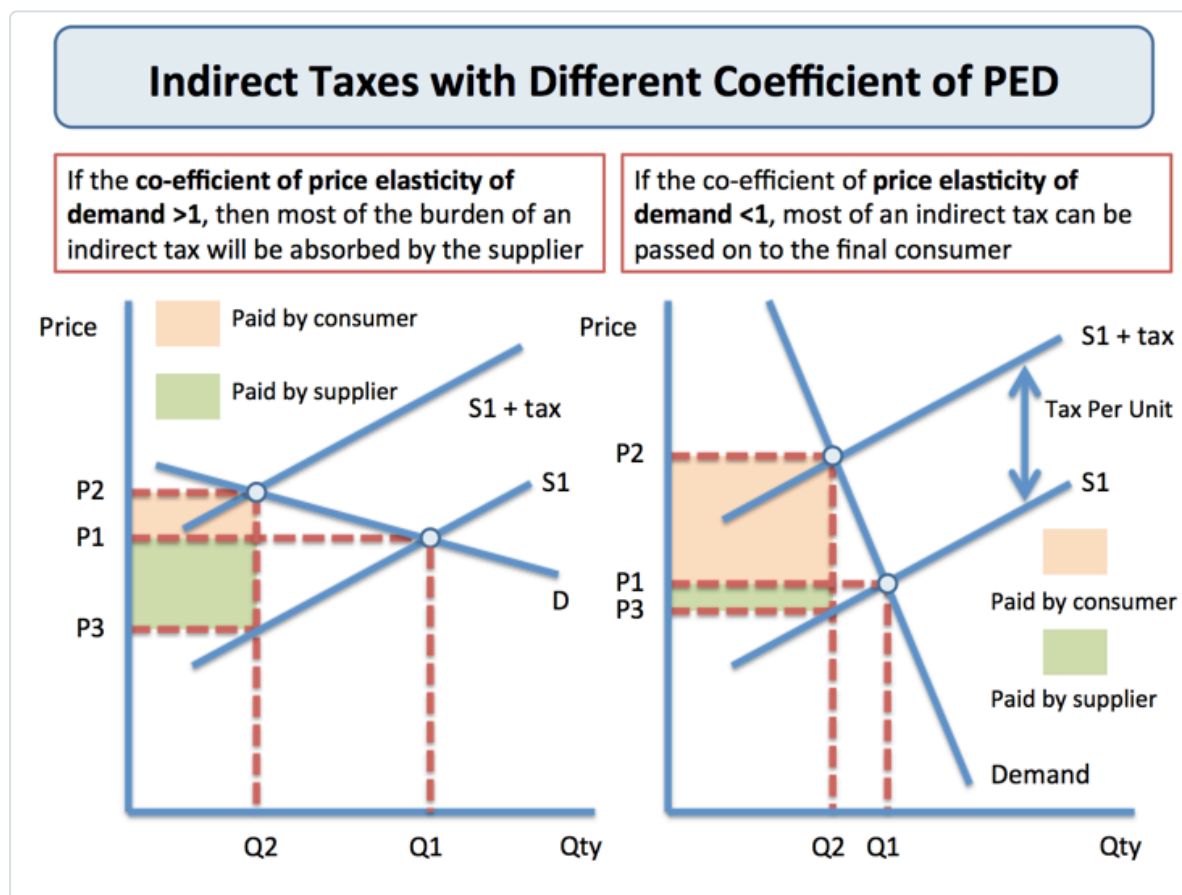
Core: a tax **drives a wedge** between the price **buyers pay** (p_b) and the price **sellers receive** (p_s): $p_b = p_s + t$. It raises the buyer price, lowers the seller price, cuts the quantity traded, and creates a **deadweight loss**. The split of the burden — **tax incidence** — depends on the **relative elasticities**, and (under perfect competition) **not** on who is legally taxed.

- **Two types of tax (deck):**
 - **Unit (specific) tax** — a fixed **amount t per unit** (per litre of petrol, per cigarette).
 - **Ad valorem tax** ("sales tax"/VAT) — a **fraction α of the price/spend** (government keeps a share of each euro spent).
- **The wedge (worked unit-tax example, $t = \text{€}3$):** with $Q_D = 23 - p_b$, $Q_S = -1 + p_s/2$, and $p_b = p_s + 3$, solving gives $p_s = 14$, $p_b = 17$, $Q = 6$. Buyers pay **€3 more** than sellers receive; quantity falls from the no-tax equilibrium.
 - **Tax revenue** = $t \times Q = 3 \times 6 = \text{€}18$. The rest of the surplus lost is **deadweight loss** (here $3 \times (7-6)/2 = \text{€}1.5$) — trades that *would* have happened but now don't.
- **Legal incidence $\neq$ economic incidence (the key result).** It **doesn't matter** whether the tax is collected from **firms** (supply shifts up by t) or **consumers** (demand shifts down by t) — the quantity, the prices actually paid/received, and the DWL are **identical**. *All that matters is that the tax drives a wedge.* (Perfect-competition assumptions make this exact: no transaction costs, perfect information.)



How to read it (tax equivalence). Both panels impose the **same \$0.55 tax** on avocados. (a) collects it from **firms** → the supply curve shifts **up** by t ; (b) collects it from **consumers** → the demand curve shifts **down** by t . In **both** cases the result is identical: **Q falls to 74**, **buyers pay 2.15**, **sellers receive 1.60**, and the wedge between them is exactly **\$0.55**. → *The economics doesn't care who writes the cheque.*

- **What does decide the split: relative elasticities.** The more **inelastic** side of the market bears **more** of the tax (it can't escape by changing quantity). The more **price-sensitive (elastic)** side is affected less on price, **more** on quantity.



How to read it (incidence & elasticity). The tax raises the supply curve to $S + \text{tax}$. The **new higher price P_2** is what **consumers** pay; **P_3** is what **suppliers** keep. The split of the rectangle between them is

the incidence:

- **Left — elastic demand ($PED > 1$):** consumers easily cut back, so firms **can't raise the price much** → the **supplier absorbs most** of the tax.
- **Right — inelastic demand ($PED < 1$):** consumers keep buying regardless, so the firm **passes most of the tax through** → the **consumer bears most** of it.
- **Policy take-aways:** (1) "**sin taxes**" on inelastic goods (cigarettes, alcohol) fall mostly on **consumers** and raise a lot of revenue with **little DWL** (quantity barely moves) — which is *why* governments like them, and why they can be **regressive** (hit the poor). (2) We **prefer taxes on inelastic bases** — they distort choices least.
- **Putting a number on the split — the pass-through formula.** The diagrams say *which* side pays more; this formula says *how much*. Each side's share of the tax is the **other side's** elasticity over the total:

TAX INCIDENCE (PASS-THROUGH)

$$\text{consumer share} = E_S \div (E_S + |E_D|) \quad \cdot \quad \text{producer share} = |E_D| \div (E_S + |E_D|)$$

- **Read it as "the other side's elasticity on top":** the **consumer** share carries **E_S** (supply's elasticity); the **producer** share carries **$|E_D|$** (demand's elasticity). Same denominator; the two shares **add to 1** (they split the whole tax — a handy check). - **Why the "other" side's elasticity?** A side pays a big share when it **can't escape** — i.e. when it's inelastic *and* the other side is elastic*. If supply is very elastic (big E_S), sellers walk away rather than absorb the tax, so it lands on **consumers** → E_S on top of the consumer share. This is the same "inelastic side bears more" rule from the diagrams, now as an equation. (Worked in [practice problem 11](#micro3-practice).) - **Subsidies are the mirror image:** $p_s = p_b + s$ — a subsidy lowers the buyer price, raises the seller price, **expands** quantity above the efficient level, and also creates a **deadweight loss** (units produced that cost more than consumers value them). Its incidence, too, depends on elasticities (inelastic demand — e.g. emergency care, primary education — → small DWL). - **Worked example** (subsidy $s = €3$): same market ($Q_D = 23 - p_b$, $Q_S = -1 + p_s/2$) but now $p_s = p_b + 3$ → solving gives $p_b = 15$, $p_s = 18$, $Q = 8$ — buyers pay **less** and sellers get **more**, and quantity **rises** above the no-subsidy level ($6 \rightarrow 8$). The subsidy costs the government $s \times Q = 3 \times 8 = €24$, part of which is **deadweight loss**.

Cases & examples

Read each as a worked use of an elasticity number (or a tax result).

- **Sugary-soda tax (Allcott et al. 2019).** Own-price elasticity ≈ -1.37 (elastic). A 5% price rise → **~7% fall** in quantity; to cut demand 15% you'd raise price **~11%**. → *Illustrates: elasticity as the **design tool for a corrective** ("sin") tax.*
- **World cocoa demand.** Inelastic (≈ -0.19 to -0.96 ; policy estimate -0.34). → *Illustrates: **inelastic demand** → **supply shocks hit price**, so bad harvests swing incomes for growers in Ghana / Côte d'Ivoire.*
- **US beer — brand vs category (Miller et al. 2021).** Product-level elasticity ≈ -4.74 , category ≈ -0.60 . → *Illustrates: **aggregation** — a single brand has close substitutes (elastic), the category doesn't (inelastic).*

- **Sweden–Denmark cross-border alcohol (Asplund et al. 2007)**. Swedes drive to Denmark for cheaper drink. Estimated for spirits: **own-price** -1.3 (elastic), **cross-price (Danish price)** $+0.3$ (substitute), **income** $+1.4$ (normal good). → *Illustrates: all three elasticities at once — own, cross, income — from real data.*
- **Electricity demand (Labandeira et al. 2017)**. ≈ -0.24 short-run vs -0.6 long-run. → *Illustrates: demand is more elastic in the long run (time to adjust equipment/habits).*

Formula sheet — quick reference

Quantity	Formula	Sign / meaning
Price elasticity of demand	$E_D = \% \Delta Q_D \div \% \Delta p$	negative ; $ E > 1$ elastic, $ E < 1$ inelastic
Point elasticity (linear)	$E_D = (dQ/dp) \cdot (p/Q)$	varies along the curve; more elastic at higher p
Constant-elastic demand	$Q = a \cdot p^{-\epsilon} \rightarrow \ln Q = \ln a + \epsilon \ln p$	elasticity = ϵ everywhere
Cross-price elasticity	$E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B$	+ substitutes, – complements
Income elasticity	$E_{\text{income}} = \% \Delta Q \div \% \Delta I$	+ normal, – inferior ; food < 1 (Engel)
Supply elasticity	$E_S = \% \Delta Q_S \div \% \Delta p$	positive
Tax wedge	$p_b = p_s + t$	buyers pay more, sellers get less; DWL
Tax revenue	$t \times Q_{\text{taxed}}$	rectangle; rest of loss = deadweight loss
Subsidy wedge	$p_s = p_b + s$	expands Q above efficient; also a DWL

Exam pointers

- Compute an elasticity from two (p, Q) points and **name** it (elastic/inelastic/unit) — the $-1/3$ example.
- Use the point-elasticity trick $E_D = (dQ/dp) (p/Q)$ on a **linear demand** and show elasticity **changes along the curve** (more elastic at higher price).
- Give the **sign rules** for cross-price (substitutes/complements) and income (normal/inferior) elasticities, with an example each.
- Solve a **unit-tax** market ($p_b = p_s + t$) for p_b , p_s , Q , compute **revenue** and **deadweight loss** — the $t = \text{€}3$ example.
- State and **explain** the two big tax results: **legal** $\neq$ **economic incidence** (doesn't matter who's taxed) and **incidence depends on relative elasticities** (inelastic side pays more); use the sin-tax application.
- Explain **why elasticity is unit-free** and why demand is **more elastic in the long run** and for narrower product categories.

Bridge. Lecture 3 turns the Chapter 2 model from *qualitative* ("the curve shifts left") into *quantitative* ("by how much, and split how between price and quantity"). Elasticity is also the hinge to later chapters: **firm pricing** (a monopolist's markup depends on E_D), **tax policy & deadweight loss** (Ch. 7), and **who gains from trade**.

★ Practice problems — the whole chapter (with worked solutions)

A set covering **every calculation** in this lecture. **Try each closed-book first**, then check against the worked solution. (*Exam is open-book, no AI — so the skill being tested is doing these fast and correctly, not memorising.*)

What you must be able to calculate (each maps to a problem below):

Skill	Formula	Problem
Own-price elasticity from two points + name it	$E_D = \% \Delta Q \div \% \Delta p$	1
Predict $\% \Delta Q$ from $\% \Delta p$ (and reverse)	$\% \Delta Q = E_D \cdot \% \Delta p$	2
Point elasticity on a linear curve (+ find the unit-elastic point)	$E_D = (dq/dp) \cdot (p/Q)$	3
Elasticity → total revenue (raise or cut price?)	$TR = p \cdot Q$	4
Constant-elastic demand	$Q = a \cdot p^{-\epsilon}$	5
Cross-price elasticity + classify	$E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B$	6
Income elasticity + classify	$E_{\text{income}} = \% \Delta Q \div \% \Delta I$	7
Supply elasticity	$E_S = \% \Delta Q_S \div \% \Delta p$	8
Unit tax: solve p_b , p_s , Q , revenue, DWL, incidence	$p_b = p_s + t$	9
Subsidy: solve p_b , p_s , Q , cost, DWL	$p_s = p_b + s$	10
Incidence from elasticities (pass-through)	$\text{consumer share} = E_S \div (E_S + E_D)$ $\text{producer share} = E_D \div (E_S + E_D)$	11
Equilibrium + a price control (Ch. 2 recap)	set $Q_D = Q_S$	12

Problems

1. A bakery raises bread from €4 → €5; weekly sales fall from 100 → 80 loaves. Find the price elasticity of demand and say whether demand is elastic, inelastic, or unit-elastic.

2. (a) The own-price elasticity of petrol is -0.3 . If the price rises **10%**, by how much does quantity demanded change? (b) A cinema wants ticket sales to rise **9%**; demand elasticity is -1.5 . What price change is needed?
3. Demand is $Q = 120 - 4p$. (a) Find E_D at $p = 10$. (b) Find E_D at $p = 20$. (c) At what price is demand **unit-elastic**?
4. Using the same demand $Q = 120 - 4p$: at $p = 10$ and at $p = 20$, should the firm **raise or cut** price to increase **total revenue**? At what price is revenue **maximised**?
5. Demand is $Q = 500 \cdot p^{-2}$. (a) What is the elasticity of demand? (b) Find quantity at $p = 5$ and $p = 10$. (c) If price rises **1%** from €5, roughly what is $\% \Delta Q$?
6. Classify each pair as substitutes, complements, or unrelated, and give E_{cross} : (a) price of **tea** **+8%** → demand for **coffee** **+4%**; (b) price of **printers** **+10%** → demand for **ink** **+6% fall**; (c) price of **umbrellas** **+10%** → demand for **coffee** **0%**.
7. Classify the good and give E_{income} : (a) income **+5%** → **restaurant meals** **+15%**; (b) income **+10%** → **bus tickets** **-4%**; (c) income **+8%** → **bread** **+2%**.
8. When the price rises from **€8** → **€10**, quantity supplied rises from **60** → **69**. Find the price elasticity of supply and classify it.
9. A market has $Q_D = 100 - 4p$ and $Q_S = -20 + 2p$. (a) Find the pre-tax equilibrium. (b) A **unit tax** $t = \text{€}6$ is imposed. Solve for p_b , p_s , and Q . (c) Find **tax revenue** and the **deadweight loss**. (d) How is the burden split between buyers and sellers, and **why**?
10. Same market ($Q_D = 100 - 4p$, $Q_S = -20 + 2p$). Now the government pays a **subsidy** $s = \text{€}6$ per unit. Solve for p_b , p_s , Q , the **cost to government**, and the **deadweight loss**.
11. At the pre-tax equilibrium of problem 9, the elasticities are $E_D = -4$ and $E_S = +2$. Use the pass-through formula to find the **share of the tax borne by consumers vs producers**, and check it against your answer to 9(d).
12. (Ch. 2 recap.) $Q_D = 80 - 2p$, $Q_S = -10 + p$. (a) Find equilibrium p^* and q^* . (b) The government sets a **price ceiling** at **€25**. Compute the resulting **shortage**.

Solutions

1. $\% \Delta Q = (80 - 100)/100 = -20\%$; $\% \Delta p = (5 - 4)/4 = +25\%$.

$$E_D = -20\% \div +25\% = -0.8$$

$|-0.8| < 1 \rightarrow$ ****inelastic**** (quantity moves less than price). * (Note: revenue actually rises — see problem 4's logic.) *

2. (a) $\% \Delta Q = E_D \cdot \% \Delta p = -0.3 \times 10\% = -3\%$ (quantity falls 3%). (b) rearrange: $\% \Delta p = \% \Delta Q \div E_D = 9\% \div (-1.5) = -6\% \rightarrow$ the cinema must **cut** price 6%.

3. Slope $dQ/dp = -4$ (constant). Use $E_D = (dQ/dp) \cdot (p/Q)$.

- (a) $p = 10 \rightarrow Q = 120 - 40 = 80 \rightarrow E_D = -4 \times 10/80 = -0.5$ (inelastic).

- (b) $p = 20 \rightarrow Q = 120 - 80 = 40 \rightarrow E_D = -4 \times 20/40 = -2$ (elastic).
- (c) unit-elastic: set $-4 \cdot (p/Q) = -1 \rightarrow Q = 4p \rightarrow 120 - 4p = 4p \rightarrow 8p = 120 \rightarrow p = 15$ ($Q = 60$). (On any straight-line demand, the unit-elastic point is the **midpoint**.)

4. Revenue $TR = p \cdot Q$.

- At $p = 10$ (inelastic, $|E| < 1$): raising price **raises** revenue. Check: $p=10 \rightarrow TR = 10 \times 80 = 800$; $p=11 \rightarrow Q = 76$, $TR = 836$. $\rightarrow$ **raise price**.
- At $p = 20$ (elastic, $|E| > 1$): cutting price **raises** revenue. Check: $p=20 \rightarrow TR = 800$; $p=19 \rightarrow Q = 44$, $TR = 836$. $\rightarrow$ **cut price**.
- **Revenue is maximised at the unit-elastic point** $p = 15$: $TR = 15 \times 60 = 900$. (Rule: $|E| < 1 \rightarrow$ raise price; $|E| > 1 \rightarrow$ cut price; max TR where $|E| = 1$.)

5. (a) For $Q = a \cdot p^\epsilon$, the elasticity is the exponent: $E_D = -2$ (constant at every price). (b) $p = 5 \rightarrow Q = 500/25 = 20$; $p = 10 \rightarrow Q = 500/100 = 5$. (c) $\% \Delta Q \approx E_D \cdot \% \Delta p = -2 \times 1\% = -2\%$.

6. $E_{cross} = \% \Delta Q_A \div \% \Delta p_B$. (a) $+4 \div +8 = +0.5 \rightarrow$ **substitutes** (tea dearer $\rightarrow$ more coffee). (b) $-6 \div +10 = -0.6 \rightarrow$ **complements** (printers dearer $\rightarrow$ less ink). (c) $0 \div +10 = 0 \rightarrow$ **unrelated**.

7. $E_{income} = \% \Delta Q \div \% \Delta I$. (a) $+15 \div +5 = +3 \rightarrow$ **normal, and a luxury** ($E > 1$). (b) $-4 \div +10 = -0.4 \rightarrow$ **inferior**. (c) $+2 \div +8 = +0.25 \rightarrow$ **normal necessity** ($0 < E < 1$; cf. Engel's law for food).

8. $\% \Delta Q_S = (69 - 60)/60 = +15\%$; $\% \Delta p = (10 - 8)/8 = +25\%$.

$$E_S = +15\% \div +25\% = +0.6$$

$0 < E_S < 1 \rightarrow$ **inelastic supply** (output hard to expand quickly).

9. (a) **Pre-tax**: set $Q_D = Q_S$: $100 - 4p = -20 + 2p \rightarrow 120 = 6p \rightarrow p^* = 20$, $Q^* = 20$.

- (b) **With $t = 6$** , use $p_b = p_s + 6$, $Q_D = 100 - 4p_b$, $Q_S = -20 + 2p_s$. Set equal: $100 - 4(p_s + 6) = -20 + 2p_s \rightarrow 76 - 4p_s = -20 + 2p_s \rightarrow 96 = 6p_s \rightarrow p_s = 16$, $p_b = 22$, $Q = 100 - 4(22) = 12$. (check $Q_S = -20 + 32 = 12$ ✓)
- (c) **Tax revenue** = $t \times Q = 6 \times 12 = \text{€}72$. **DWL** = $\frac{1}{2} \times t \times (Q^* - Q_{tax}) = \frac{1}{2} \times 6 \times (20 - 12) = \text{€}24$.
- (d) Buyers bear $p_b - p^* = 22 - 20 = \text{€}2$; sellers bear $p^* - p_s = 20 - 16 = \text{€}4$ (they sum to $t = 6$). **Sellers bear more because supply is the more inelastic side** (slope 2 vs demand's 4 $\rightarrow$ demand is more elastic, so it escapes more of the tax onto quantity).

10. **With $s = 6$** , use $p_s = p_b + 6$: $100 - 4p_b = -20 + 2(p_b + 6) \rightarrow 100 - 4p_b = -8 + 2p_b \rightarrow 108 = 6p_b \rightarrow p_b = 18$, $p_s = 24$, $Q = 100 - 4(18) = 28$. (check $Q_S = -20 + 48 = 28$ ✓) — buyers pay **less** ($18 < 20$), sellers get **more** ($24 > 20$), and **Q rises** ($20 \rightarrow 28$). **Cost to government** = $s \times Q = 6 \times 28 = \text{€}168$. **DWL** = $\frac{1}{2} \times s \times (Q_{sub} - Q^*) = \frac{1}{2} \times 6 \times (28 - 20) = \text{€}24$ (over-production: units that cost more than they're worth).

11. Pass-through shares:

INCIDENCE FROM ELASTICITIES

$$\text{consumer share} = E_S \div (E_S + |E_D|) \quad \cdot \quad \text{producer share} = |E_D| \div (E_S + |E_D|)$$

With $E_S = 2$ and $|E_D| = 4$: **consumer share** $= 2 \div (2+4) = 1/3$, **producer share** $= 4 \div 6 = 2/3$. Of the €6 tax that's **€2 on buyers**, **€4 on sellers** — exactly matching 9(d). ✓ (Rule: the **more inelastic** side pays the bigger share.)

12. (a) $80 - 2p = -10 + p \rightarrow 90 = 3p \rightarrow p^* = 30, q^* = 20$. (b) At the ceiling $p = 25$: $Q_D = 80 - 50 = 30, Q_S = -10 + 25 = 15 \rightarrow \text{shortage} = Q_D - Q_S = 30 - 15 = 15 \text{ units}$.

★ Fun facts & memorable details (Lecture 3)

Sticky bits from Elasticities & Taxes.

- **It doesn't matter who you tax.** In a competitive market the burden of a tax is **identical** whether the government bills the shop or the shopper — "all that matters is that the tax drives a wedge." (This is why arguing about *who* a tax is "on" often misses the point.)
- **Why cigarette & alcohol taxes raise so much money:** demand is **inelastic**, so people keep buying — the tax is **passed to consumers** with little drop in quantity (and is often **regressive**).
- **The soda-tax number:** the own-price elasticity of sugary drinks (≈ -1.37) is a real input to real policy — a 5% price rise cuts consumption $\sim 7\%$.
- **Engel's law (1857):** the **richer** you get, the **smaller** the share of income you spend on **food** — one of the oldest empirical regularities in economics, and still used to gauge living standards.
- **Pharma = "defensive stock":** demand for medicine barely moves with income (income-inelastic), so drug companies hold up in recessions.
- **Elasticity has a time dimension:** when petrol jumps, you can't do much *this week* (inelastic), but over years you buy a smaller car and move closer to work (elastic) — the "short-run vs long-run" split.
- **Brand vs category:** demand for *a* beer is wildly elastic (≈ -4.7 — switch brands!), but demand for *beer* is inelastic (≈ -0.6) — same drink, opposite elasticity, just different zoom level.
- **Swedes really do drive to Denmark for cheap booze** — enough that economists measured the cross-border **cross-price elasticity** of spirits ($+0.3$) from the data (Asplund et al. 2007).

Lecture 4: Consumer Choice (Chapter 5)

Required reading: Friberg, Chapter 5 — Consumer choice (§5.1 modelling choice • §5.2 preferences & utility • §5.3 budget constraints • §5.4 utility maximization • §5.5 discussion of the framework • §5.6 behavioral economics • §5.7 appendix: solving for the optimal basket with Lagrange). *Lecturer: Luigi Butera* — *deck "Lecture 4: Consumer choice."* (Numbering note: this is the course's **4th lecture** but the textbook's **Chapter 5**; the deck's own "4.1–4.5" roadmap just re-numbers the chapter's sections.)

Theme of the lecture (Butera — from the deck). So far (Lectures 2–3) we took **demand curves as given** and pushed them around. Now we **build demand from the ground up**: *where do preferences come from, and how does a rational consumer choose under scarcity?* The deck's answer is a **three-stage machine** — **preferences → utility → budget constraint → constrained optimization** — that will (next chapter)

generate the demand curve. The deck's five stops: 4.1 preferences · 4.2 utility · 4.3 budget constraints · 4.4 constrained choice · 4.5 behavioral economics.

★ **The heart of the lecture:** a consumer **maximizes utility** subject to a **budget constraint**. We (1) describe what she **likes** with well-behaved **preferences** (complete + transitive), (2) quantify them with a **utility function** whose slope-of-indifference-curve is the **MRS** = how much of one good she'll trade for the other *and stay equally happy*, (3) describe what she **can afford** with a **budget line** whose slope is the **MRT** = p_F/p_E = how the market lets her trade the goods. The **optimum** is where the highest reachable indifference curve is **tangent** to the budget line — i.e. where **MRS** = **MRT** ($MU_F/MU_E = p_F/p_E$): the rate she's *willing* to trade equals the rate the market *lets* her trade.

The red thread. Every piece answers one of three questions: *what do I want?* (preferences → utility → MRS), *what can I afford?* (budget line → MRT), and *so what do I pick?* (tangency: MRS = MRT). Get those three and you can predict how choices shift when **prices** or **income** change — which is exactly a **demand curve** (next chapter). §5.6 then asks: *when does this rational machine break down?* (behavioral economics).

Key concepts / "modes" to use

- **Bundle** — a list of quantities of each good (e.g. "4 Entertainment, 2 Food"). Choice = ranking bundles.
- **Preferences & the rationality axioms** — **completeness** (can rank any two bundles) · **transitivity** (rankings "hold in transit") · **monotonicity** / "more is better" (convenient add-on).
- **Utility function** $U(F, E)$ — assigns a number to each bundle so that **higher** = **preferred**. **Ordinal**, not cardinal (only the *ranking* means anything).
- **Marginal utility (MU)** — extra utility from one more unit of a good, holding the other fixed = the **partial derivative**; typically **diminishing**.
- **Indifference curve** — all bundles giving the *same* utility; **5 properties**; three shapes (**perfect substitutes**, **imperfect substitutes**, **perfect complements**).
- **Marginal rate of substitution (MRS)** — slope of the indifference curve = MU_F / MU_E = how much E you'll give up for one more F at equal utility.
- **Budget constraint / line** — $I = p_F \cdot F + p_E \cdot E$; the **opportunity set** is everything on or under it.
- **Marginal rate of transformation (MRT)** — slope of the budget line = p_F / p_E = how the *market* lets you swap the goods.
- **Consumer optimum** — highest indifference curve tangent to the budget line: **MRS** = **MRT**, i.e. $MU_F/MU_E = p_F/p_E$ (equivalently, equal marginal utility *per euro*: $MU_F/p_F = MU_E/p_E$).
- **Interior vs corner solution** — tangency inside the axes vs "spend everything on one good" (perfect substitutes).
- **Behavioral economics** — systematic deviations: **social preferences**, **lack of self-control**, **cognitive biases** (loss aversion, endowment effect, framing, overconfidence).

Section-by-section main points

§5.1 How can we model choice? — the three ingredients (deck)

Core: we model an individual who **faces constraints** (income, time) and **does as well as possible** given them. Sounds almost tautological — the whole chapter is about *putting structure* on "does as well as possible." The deck lists the **ingredients we need**:

- A language to say "I like this more than that" → **preferences** (§5.2).
- **Assumptions** that make those preferences *rational* → the **axioms** (completeness, transitivity).
- A tool to quantify wellbeing → the **utility function** (§5.2.1).
- A tool to quantify "you can't have it all" → the **budget constraint** (§5.3).
- **Bringing it together** → **constrained optimization** (§5.4).
- **Three premises of the whole build (deck + §5.1):** (1) we know and have stable preferences, (2) we face constraints (time, money), (3) we try to do as well as possible given them.

§5.2 Preferences and the rationality axioms (deck slides 6–7)

Core: choice is modelled as **ranking bundles**. For the ranking to give a consistent model we assume it obeys two **essential** axioms (plus one convenient one). The deck's notation for the ranking:

Symbol	Reads	Meaning
$a \succ b$	"a is strictly preferred to b"	a is <i>better</i> than b
$a \succeq b$	"a is weakly preferred to b"	a is <i>at least as good</i> as b
$a \sim b$	" indifferent between a and b"	a and b are <i>equally good</i>

- **Completeness (essential):** the consumer can **rank any two bundles** — for any A, B she can say $A \succ B$, $B \succ A$, or $A \sim B$. (No "I can't compare these.")
- **Transitivity (essential):** rankings "**hold in transit**" — if $A \succ B$ and $B \succ C$, then $A \succ C$. This is what lets us predict choice in a *chain* of comparisons (and stops preferences from cycling).
- **Monotonicity / "more is better" (convenient, not essential):** all else equal, **more of a good is better**. This is what makes bundles *farther from the origin* preferred. (*Things you want less of — pollution, noise — are relabelled "bads," and we redefine the good as their absence.*)
- **Why the axioms matter:** completeness + transitivity are *exactly* the conditions under which preferences can be represented by a **utility function** (next section). No axioms → no **utility function** → no model.

§5.2.1 Utility & utility functions (deck slides 8–9)

Core: if preferences are **complete and transitive**, we can represent them with a **utility function** $U(\cdot)$ that assigns each bundle a **number**, higher for more-preferred bundles. Neoclassical economics inherits this from the **utilitarian** tradition (J.S. Mill: "actions are right in proportion as they tend to promote happiness").

UTILITY REPRESENTS PREFERENCES

$$a \succ b \iff U(a) > U(b)$$

Reads: the utility function just **translates the ranking into real numbers** — bundle a is preferred to b if **and only if** a scores a higher utility number. (This is the **von Neumann–Morgenstern** representation the deck cites.) The function *orders* preferences; it does not measure "happiness" in any absolute unit.

- **Ordinal, not cardinal (the crucial caveat):** only the **ranking** matters, **not** the size of the gap. $U = 8$ vs $U = 4$ means the first is **preferred** — it does **not** mean "twice as good." (*Deck: "9 > 1, but do '1' and '9' mean something meaningful?" — no. Utility is unobservable.*)
- **General vs specific form:** general $U = U(F, E)$; a specific example the textbook uses is $U = \sqrt{EF}$. **Worked:** at $E=2, F=8$, $U = \sqrt{(2 \cdot 8)} = \sqrt{16} = 4$. Bundles with the **same** product $E \cdot F = 16$ (e.g. $E=4, F=4$ or $E=8, F=2$) sit on the **same** indifference curve; $E=5, F=5$ gives $U = 5 \rightarrow$ a **higher** curve.
- **Common functional forms (deck slide 12):**
 - **Cobb-Douglas:** $U = F^a \cdot E^b$ — the workhorse; convex, imperfect substitutes.
 - **Perfect substitutes:** $U = a \cdot F + b \cdot E$ — straight-line indifference curves.
 - (*Perfect complements:* $U = \min(F, E)$ — *L-shaped.*)

★ Marginal utility & diminishing marginal utility (deck slides 10, 16)

Core: marginal utility (MU) is the extra utility from **one more (infinitesimal) unit** of a good, holding the others fixed — i.e. the **partial derivative** of U . It is usually **diminishing** (the 1st slice of pizza beats the 7th).

MARGINAL UTILITY

$$MU_F = \partial U(F, E) / \partial F \quad \cdot \quad MU_E = \partial U(F, E) / \partial E$$

Reads: **how much total utility rises per extra unit of one good, other goods held constant** — $\partial U / \partial F$ = the partial derivative w.r.t. Food • **diminishing MU** = this derivative *shrinks* as you get more of the good.

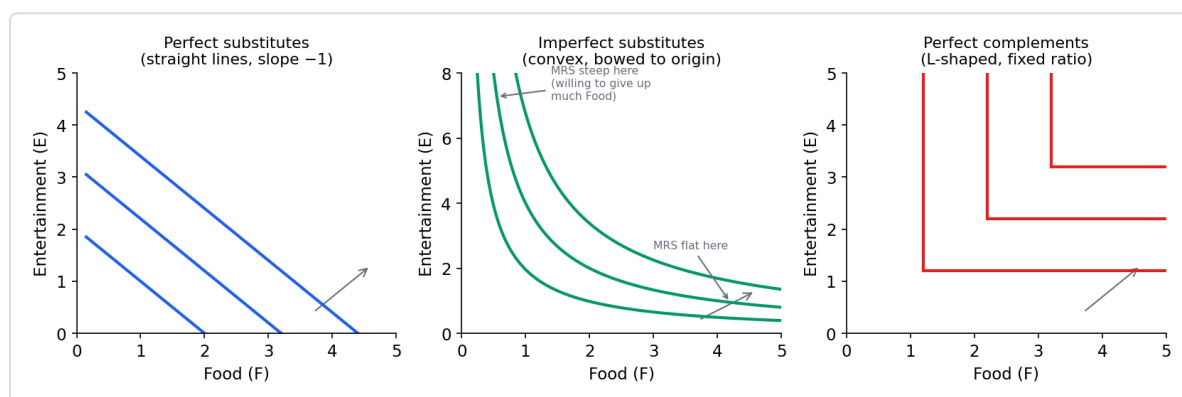
- **The deck's three consumers (utility over one good):** for each extra apple —
 - **Bob:** each apple adds the **same** utility $\rightarrow$ **constant** MU (linear U).
 - **Linda:** each apple adds a bit **more** $\rightarrow$ **increasing** MU (convex U). (*unusual*)
 - **Mike:** each apple adds a bit **less** $\rightarrow$ **diminishing** MU (concave U). (*the normal case*)
- **Worked example ($U = \sqrt{EF} = E^{\{0.5\}} F^{\{0.5\}}$):** $MU_F = \partial U / \partial F = 0.5 \cdot E^{\{0.5\}} \cdot F^{\{-0.5\}} = 0.5 \cdot \sqrt{E/F}$. At $E=2, F=8$: $MU_F = 0.5 \cdot \sqrt{(2/8)} = 0.5 \cdot 0.5 = 0.25$. As F rises (holding E), $\sqrt{E/F}$ falls $\rightarrow$ **MU_F falls** $\rightarrow$ diminishing marginal utility. ✓
- **Why it matters:** diminishing MU is *why* indifference curves are **bowed toward the origin** (convex) and *why* the **MRS falls** as you move down a curve — the next two ideas.

Indifference curves — 5 properties & the 3 types (deck slides 13–14)

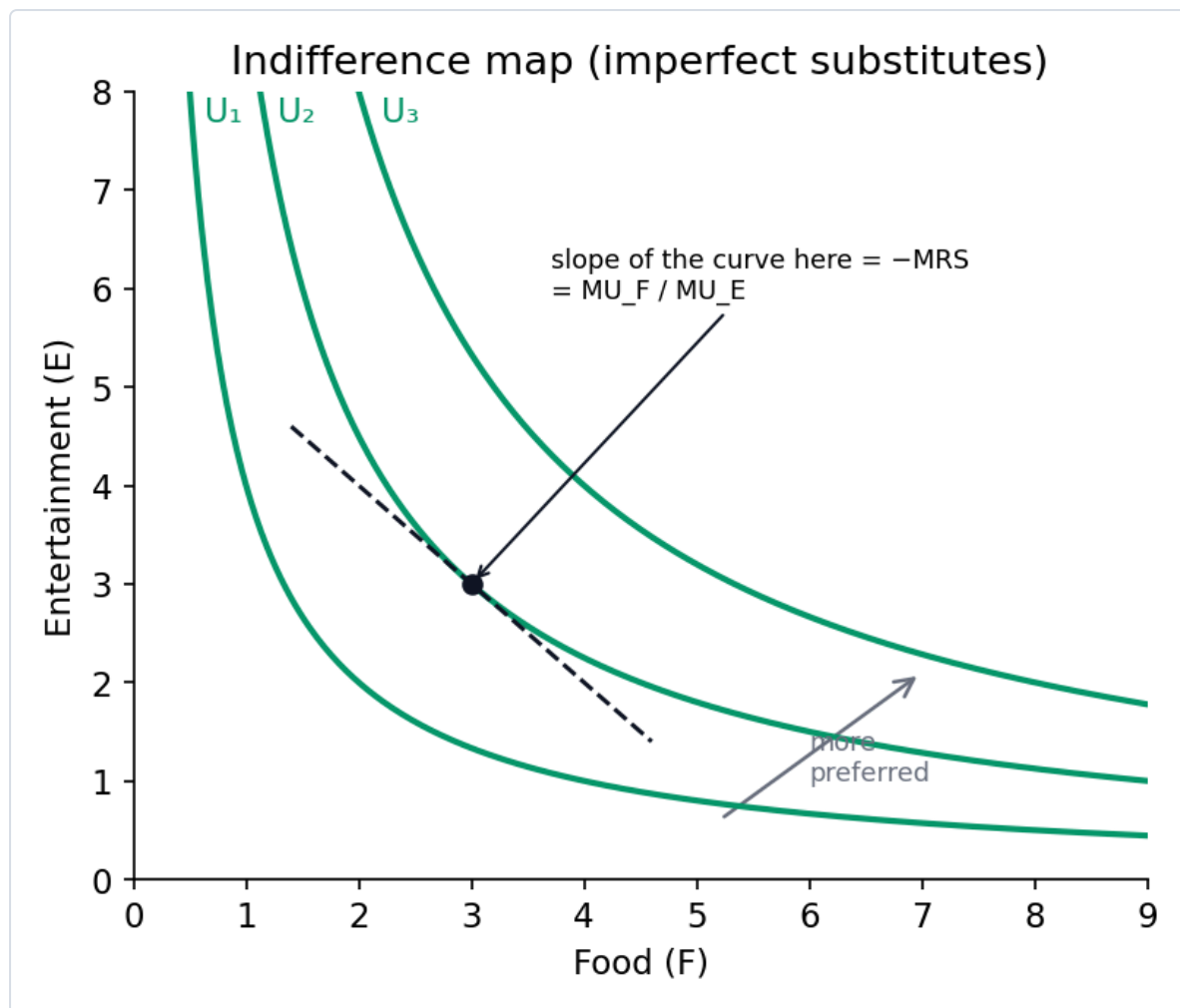
Core: an indifference curve connects **all bundles the consumer views as equally desirable**. It's the level-set of the utility function (fix U at one number). The deck's **five properties**:

#	Property	Why
1	Curves farther from the origin are preferred	"more is better" (monotonicity)
2	Every bundle lies on some <u>indifference curve</u>	completeness
3	Indifference curves cannot cross	transitivity (crossing → contradiction)
4	They slope downward	to keep utility fixed, more of one good ⇒ less of the other
5	They cannot be "thick"	a thick band would contain a strictly-better bundle (monotonicity)

- (Utility is read **ordinally**: the labels on the curves rank them; the gaps between labels are meaningless.)
- Three shapes of preference (textbook §5.2):



- **Perfect substitutes** — tradeable at a **constant rate** → **straight-line** curves (constant slope). *Example: one €20 note ~four €5 notes, always.* $U = aF + bE$.
- **Imperfect substitutes (the normal case)** — willing to trade, but the **rate depends on how much of each you already have** → **convex** curves, "bowed toward the origin" (**strictly convex** preferences). *When you have lots of chips and no drink, you'll give up many chips for one drink; when you have few chips, you won't.* $U = \sqrt{EF}$, Cobb-Douglas.
- **Perfect complements** — consumed in a **fixed ratio** → **L-shaped** curves. *Left & right shoes: a 2nd right shoe alone adds nothing.* $U = \min(F, E)$.



How to read it (indifference map). Each green curve is one utility level; **northeast = more preferred** (property 1). The dashed tangent at the marked bundle has slope $-MRS$: it gets **flatter** as you move right (more Food, less willing to give up Food) — that's diminishing MRS, the topic below.

★ Marginal rate of substitution (MRS) (deck slides 17–19)

Core: the **MRS** is the slope of the indifference curve — the maximum amount of one good the consumer will give up to get one more unit of the other, staying equally happy. Because utility is constant along the curve ($dU = 0$), it equals the **ratio of marginal utilities**.

MARGINAL RATE OF SUBSTITUTION

$$MRS = -dE/dF = MU_F / MU_E \quad (\text{good on the x-axis over good on the y-axis})$$

Reads: how much E you'll trade for one more F and stay on the same curve — derive it by **totally differentiating** U and setting $dU = 0$: $0 = MU_F \cdot dF + MU_E \cdot dE \rightarrow -dE/dF = MU_F / MU_E$. MU_F = marginal utility of the x-axis good (Food) · MU_E = marginal utility of the y-axis good (Entertainment) · the ratio = the curve's steepness at that point.

- **Diminishing MRS (why the curve bows):** as you move **down-right** along the curve (more F, less E), MU_F falls and MU_E rises, so the ratio MU_F / MU_E **shrinks** — the curve **flattens**. "When I have relatively less pizza I'll give up more burritos for a slice than when I already have lots of pizza."

- **Worked example ($U = \sqrt{EF}$):** $MU_F = 0.5\sqrt{E/F}$, $MU_E = 0.5\sqrt{F/E}$, so $MRS = MU_F/MU_E = (0.5\sqrt{E/F})/(0.5\sqrt{F/E}) = E/F$. At bundle $E=2, F=8$: $MRS = 2/8 = 0.25 \rightarrow$ she'll give up only **0.25 units of E** for one more F (she already has lots of F). At $E=8, F=2$: $MRS = 8/2 = 4 \rightarrow$ now she'll give up **4 units of E** for one more F. *Same curve, opposite ends $\rightarrow$ diminishing MRS.* ✓

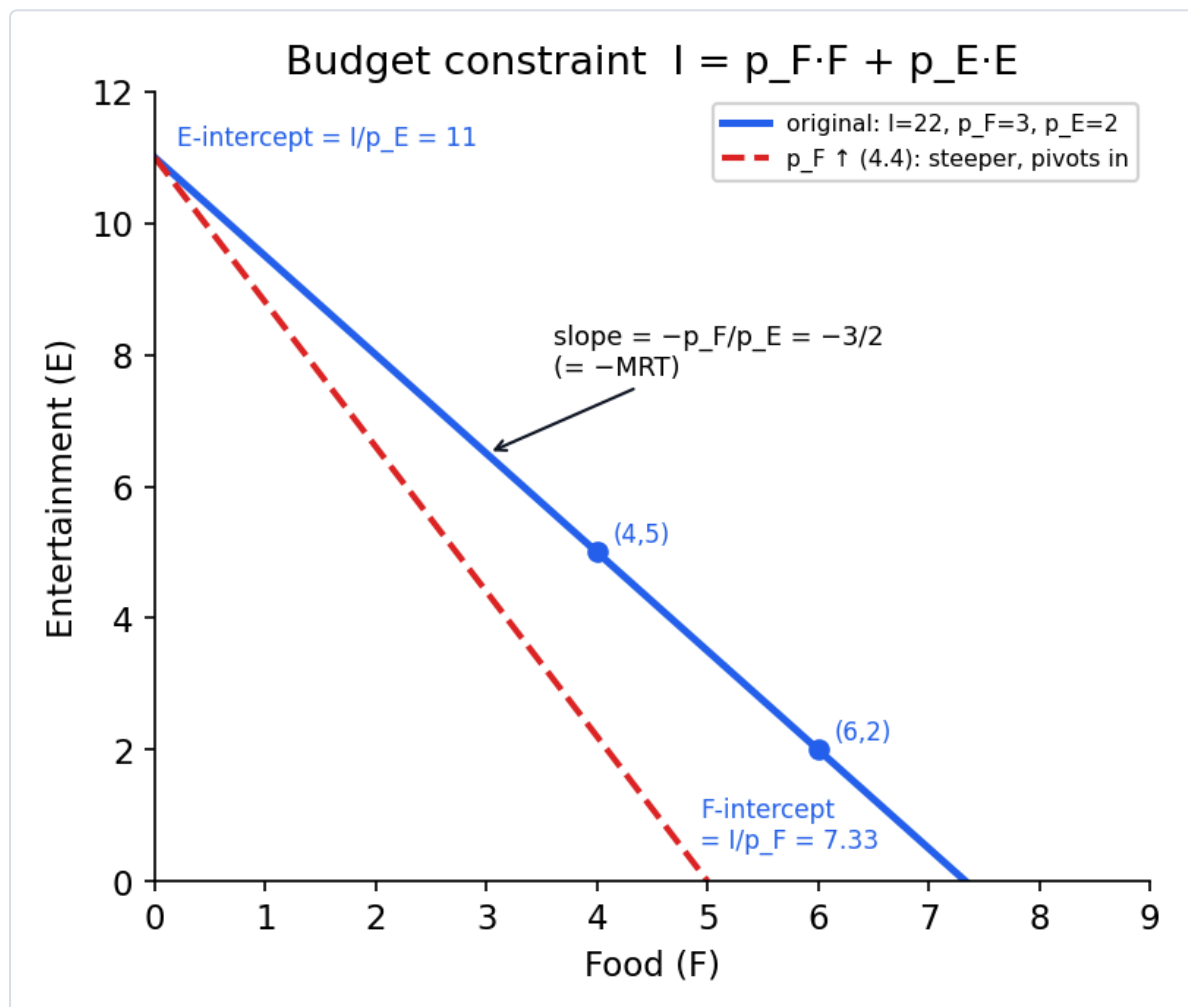
§5.3 Budget constraint & the MRT (deck slides 21–22)

Core: the **budget constraint** operationalizes "you can't have it all": given prices and income I , total spending can't exceed income. With two goods it's a **straight line** (the **budget line**); everything on or below it is the **opportunity set**.

BUDGET CONSTRAINT & LINE

$$I = p_F F + p_E E \implies E = I/p_E - (p_F/p_E) \cdot F$$

Reads: income = spending on Food + spending on Entertainment; rearranged for graphing, E-intercept = I/p_E (all income on E), slope = $-p_F/p_E$, F-intercept = I/p_F (all income on F).



- **Worked example (deck: $I=22, p_F=3, p_E=2$):** E-intercept = $22/2 = 11$; slope = $-3/2$; F-intercept = $22/3 \approx 7.33$. Bundle $(F=4, E=5)$ is affordable: $3 \cdot 4 + 2 \cdot 5 = 22$ ✓. So is $(F=6, E=2)$: $3 \cdot 6 + 2 \cdot 2 = 22$ ✓.

- **MRT — marginal rate of transformation** = the **slope of the budget line** = how the **market** lets you swap goods:

MARGINAL RATE OF TRANSFORMATION

$$MRT = p_F / p_E \quad (\text{the relative price = the market "exchange rate" between the goods})$$

Reads: give up one F, the market gives you p_F/p_E units of E. With $p_F=3$, $p_E=2$, $MRT = 1.5$: one less Food frees €3, which buys 1.5 Entertainment. (*MRS = what you're willing to trade; MRT = what the market lets you trade.*)

- **Comparative statics of the line (deck):**
 - **Income change** → **parallel shift** (slope unchanged; both intercepts move).
 - **Price change** → **pivot** (steepness changes). A higher $p_F \Rightarrow$ **steeper**, pivoting **inward** around the E-intercept (see red dashed line); a lower $p_F \Rightarrow$ **flatter**.

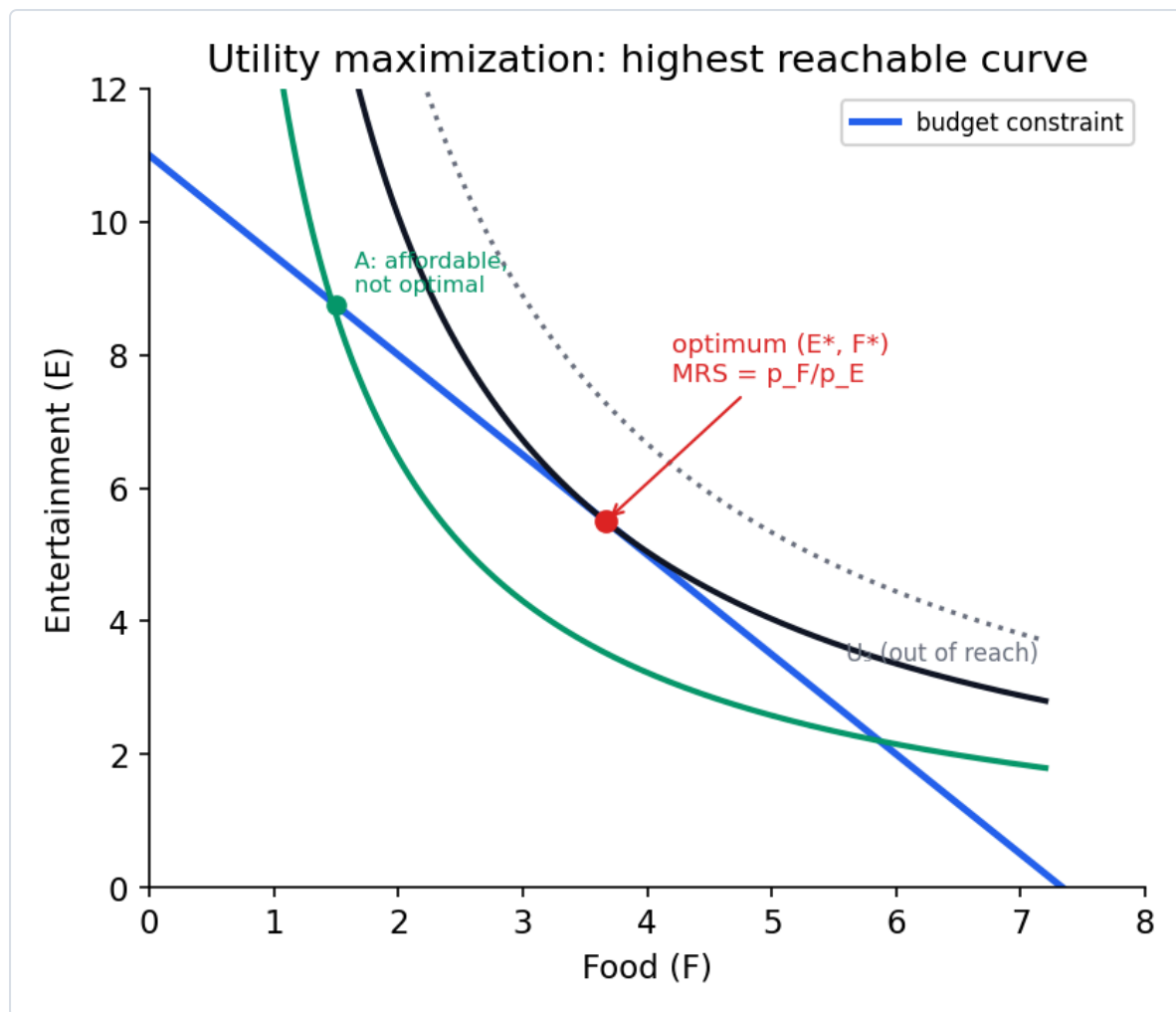
★ §5.4 The optimum: maximize utility subject to the budget (deck slide 23)

Core: the consumer solves **max $U(F,E)$ subject to $p_F \cdot F + p_E \cdot E \leq I$** . Graphically: find the **highest indifference curve** that still touches the budget line — that's the **tangency point**. At the tangency, the slope of the indifference curve (**MRS**) equals the slope of the budget line (**MRT**).

CONSUMER OPTIMUM (TANGENCY)

$$MRS = MRT \iff MU_F/MU_E = p_F/p_E \iff MU_F/p_F = MU_E/p_E$$

Reads: at the optimum, **the rate you're willing to trade = the rate the market lets you trade**. The last form is the "**equal marginal utility per euro**" (bang-for-the-buck) rule: **the last euro spent on Food yields the same extra utility as the last euro spent on Entertainment** — otherwise you'd reallocate toward the higher-MU-per-euro good.



How to read it (utility maximization). Point A is *affordable* (on the budget line) but **not optimal**: the curve through A is steeper than the line ($MRS > MRT$), so she values Food more than the market charges → she should **buy more Food, less Entertainment**, sliding down to the tangency. The **red optimum** is the best reachable bundle; the dotted curve U_3 would be better but her income can't reach it.

- **Why A isn't optimal (the deck's logic):** at A, $MU_F/MU_E > p_F/p_E$ — food is "cheap relative to how much you value it." Reallocate toward food until the two ratios are equal.
- **Worked example ($U = \sqrt{EF}$), $I=22$, $p_F=3$, $p_E=2$):** $MRS = E/F$. Set $MRS = MRT$: $E/F = 3/2$ → $E = 1.5F$. Plug into the budget $3F + 2E = 22$: $3F + 2(1.5F) = 22$ → $6F = 22$ → $F^* = 3.67$, $E^* = 5.5$. (Check: $3 \cdot 3.67 + 2 \cdot 5.5 = 11 + 11 = 22$ ✓ — matches the red dot in the figure.)

Comparative statics · interior vs corner · the Cobb-Douglas shortcut (deck slide 24, §5.7)

Core: once you can find the optimum, you can ask **how it moves** when a price or income changes — and that movement, traced out, **is the demand curve** (Ch. 6). Watch for **corner solutions**.

- **Interior vs corner solution (deck):**
 - **Interior solution** — the tangency lies **strictly inside** the axes (buy *some* of both goods). Requires **convex** (imperfect-substitute) preferences.
 - **Corner solution** — the optimum is **on an axis** (buy *only one* good). Happens with **perfect substitutes**: if $MRS \neq MRT$ everywhere, you spend **everything** on whichever good gives more

utility per euro (MU/p). Example: if $U = 2F + E$ and $p_F = p_E$, Food gives twice the bang per euro $\rightarrow$ buy only Food.

- **Perfect complements** — always a corner-of-the-L: buy the goods in their **fixed ratio** regardless of the price ratio (solve the ratio + the budget line together).
- **The Cobb-Douglas shortcut (worth memorising for exams):** for $U = F^a \cdot E^b$, the tangency condition solves in closed form — the consumer **spends a constant share of income on each good**:

COBB-DOUGLAS OPTIMAL DEMANDS

$$F^* = \left[\frac{a}{a+b} \right] \cdot I/p_F \quad \cdot \quad E^* = \left[\frac{b}{a+b} \right] \cdot I/p_E$$

Reads: spend the fraction $a/(a+b)$ of income on Food and $b/(a+b)$ on Entertainment, then divide by each price to get quantities. The exponents are the expenditure shares. **Worked ($U = F^{0.5}E^{0.5}$)**, so $a=b=0.5$, $I=22$, $p_F=3$, $p_E=2$: $F^* = 0.5 \cdot 22/3 = 3.67$, $E^* = 0.5 \cdot 22/2 = 5.5$ — same answer as the tangency method above, in one line. (Notice each good's demand **doesn't depend on the other good's price** — a special Cobb-Douglas feature.)

- **The appendix method (§5.7) — Lagrange:** for a general U , combine objective + constraint into $\mathcal{L} = U(F,E) + \lambda(I - p_F F - p_E E)$, set the three partials to zero, and the first two give back $MU_F/MU_E = p_F/p_E$ — the same tangency condition. λ = the **marginal utility of income** ("how much extra utility one more euro of income buys"). *You won't need Lagrange for the exam, but know that it reproduces $MRS = MRT$.*
- **The payoff (Ch. 6 foreshadow):** change p_F , re-solve the optimum, and record the new F^* — repeat, and the (p_F, F^*) pairs **trace out the demand curve**. *This is why we built all this machinery: choice $\rightarrow$ demand.*

§5.5–5.6 Discussion of the framework & behavioral economics

Core: the rational-choice model is powerful (Becker: "maximizing behavior, market equilibrium, and stable preferences, used relentlessly and unflinchingly") — but it's a **model**. Two clarifications, then the **deviations**.

- **Two common misconceptions (§5.5):**
 - "Economics says be a selfish materialist." **No** — the framework maximizes **utility**, not **goods**. Giving to charity, choosing a low-paid academic job you love, buying organic — all perfectly rational if that's your preference.
 - "It only works for apples-vs-oranges." Actually **Gary Becker** extended it to fertility, marriage, crime, discrimination: assume **stable preferences** and let **costs/benefits differ** across contexts (e.g. why fertility differs across countries — Table 5.2 — need not be different *tastes*).
- **Behavioral economics (§5.6) — three deviations from the simple model:**

Deviation	What it is	Signature examples
Social preferences	We care about others — fairness, status, altruism, spite. <i>Not irrational</i> — just richer preferences.	Ultimatum game : proposers offer ~40% (not ~0%), responders reject ~16% of low offers to punish unfairness.
Lack of self-control	We over-weight the present — a myopic " <i>doer</i> " vs a " <i>planner</i> " (Thaler–Shefrin).	Smoking, over-spending, under-saving; the marshmallow test; policy responses = mandatory pensions, alcohol limits.
Cognitive limitations & biases	Systematic (not random) errors; bounded rationality , heuristics.	Loss aversion / status-quo bias / endowment effect (Cornell coffee-mug study: owners value the mug 2–4× more); framing (600-lives problem); overconfidence (68% of entrepreneurs think they'll beat the odds).

- **Random error vs bias (deck's dartboard)**: *random* misses scatter around the bull's-eye (people are right *on average*); a **behavioral bias** is a **systematic** skew off-centre (Kahneman & Tversky, *Thinking, Fast and Slow*).
- **Final word (§5.6.5)**: the rational model stays the **benchmark** — you need it *even to define* a "deviation." Behavioral economics tells you **when** the benchmark will mislead (e.g. demand for addictive painkillers vs baking soda).

Cases & examples

- **Why did shrimp consumption triple? (Freakonomics / Levitt) — §5.4.1**. Non-economists guessed **demand shifts** (health fads, movies, trendy chefs). The choice framework points first to **prices and income**: the **relative price of shrimp fell** — that alone moves the optimum along/across the budget line. *Lesson: shifts in the budget constraint, not just tastes, drive choices.*
- **Same preferences, different choices: US pizza vs French duck confit (Dubois et al. 2014) — Fig 5.8**. Give an American and a French consumer the **identical** indifference map. Pizza is **cheaper in the US**, confit **cheaper in France** → different **budget-line slopes** → the American picks more pizza, the French more confit. *Different choices need not mean different tastes — just different relative prices.*
- **The ultimatum game — fairness is real (Güth 1982; Henrich 2001)**. Split €10; responder can veto (then both get €0). Pure self-interest predicts "offer €0.10, accept." Instead proposers offer **~40%**, and low offers get **rejected** — even across 15 small-scale societies worldwide (offers never below ~25%). *Evidence for social preferences.*
- **The Cornell coffee-mug experiment — the endowment effect (Kahneman et al. 1990)**. Mugs handed out at random; owners then valued them **2–4× higher** than non-owners would pay. Ownership shouldn't matter to a rational valuation — but a **reference point** does. *Loss aversion / status-quo bias in one image.*
- **Food stamps or cash? (deck slide 25)**. ~50% of US kids use food stamps before age 20. In the standard model, **cash is weakly better** (it relaxes the budget line *more* — you can buy food *or* anything else), so an in-kind transfer can only tie or lose. Yet food stamps persist — because of **paternalism / self-control** concerns and political constraints. *A perfect budget-constraint application with a behavioral twist.*

- **Fertility rates: tastes or costs? (Becker; Table 5.2).** South Korea 0.97 kids/woman vs Mali 6.14. A Becker-style reading: **same preferences**, different **costs/benefits** of children (childcare compatibility with careers, old-age support, child mortality, pensions). *Rational choice reaches far beyond "goods."*

Formula sheet (Consumer Choice)

Concept	Formula	In words
Utility represents preferences	$a > b \Leftrightarrow U(a) > U(b)$	numbers rank bundles (ordinal)
Marginal utility	$MU_F = \partial U / \partial F$	extra utility per extra unit; usually diminishing
MRS (slope of indifference curve)	$MRS = MU_F / MU_E$	how much E you'll trade for one more F, same utility
Budget line	$I = p_F \cdot F + p_E \cdot E \rightarrow E = I/p_E - (p_F/p_E)F$	what you can afford; slope $-p_F/p_E$
MRT (slope of budget line)	$MRT = p_F / p_E$	how the market lets you trade F for E
<u>Consumer optimum</u>	$MRS = MRT \Leftrightarrow MU_F/MU_E = p_F/p_E \Leftrightarrow MU_F/p_F = MU_E/p_E$	tangency = equal MU per euro
Cobb-Douglas demands ($U = F^a E^b$)	$F^* = a/(a+b) \cdot I/p_F, E^* = b/(a+b) \cdot I/p_E$	spend constant income shares

Key terms

Term	One-line meaning
Bundle	a specific quantity of each good
Completeness	can rank <i>any</i> two bundles
Transitivity	$A > B, B > C \Rightarrow A > C$ (rankings "hold in transit")
Monotonicity	"more is better" — farther-out curves preferred
Utility function	numbers on bundles preserving the ranking (ordinal)
Marginal utility (MU)	$\partial U / \partial (\text{good})$; extra utility from one more unit
Indifference curve	all bundles of equal utility
MRS	slope of indifference curve = MU_F / MU_E
Budget constraint / line	$I = p_F F + p_E E$; the opportunity set
MRT	slope of budget line = p_F / p_E
Consumer optimum	highest reachable curve tangent to the budget line
Interior / corner solution	some of both goods / all of one good
Behavioral economics	systematic deviations from the rational model
Loss aversion / endowment effect	losses hurt more than equal gains; owning raises valuation

Exam pointers

- **Draw the machine, then solve it.** For "find the optimal bundle": (1) write $MRS = MU_F / MU_E$, (2) set $MRS = p_F / p_E$, (3) combine with the **budget line** $I = p_F F + p_E E$, (4) solve the two equations. *Two equations, two unknowns — always.*
- **Cobb-Douglas → use the shortcut.** If $U = F^a E^b$, jump straight to $F^* = a / (a+b) \cdot I / p_F$. Saves time; derive it once so you trust it.
- **Check the second good.** After finding one quantity, plug back into the budget line for the other, and **verify it satisfies the constraint** ($p_F F^* + p_E E^* = I$).
- **MRS = MU of the x-axis good over MU of the y-axis good.** Getting the ratio upside-down is the #1 error. If Food is on the x-axis, $MRS = MU_F / MU_E$ and the tangency is $MU_F / MU_E = p_F / p_E$.
- **Corner solutions:** perfect substitutes → compare MU_F / p_F vs MU_E / p_E and spend everything on the winner. Perfect complements → ignore the price ratio, use the **fixed ratio + budget line**.
- **Comparative statics language:** income change = **parallel shift**; own-price change = **pivot**. Re-solving the optimum after a price change is how you **derive a demand curve** (Ch. 6).
- **Ordinal, not cardinal:** never say "twice as happy." Utility numbers only **rank**.
- **Behavioral:** be able to name the bias from a scenario (endowment effect, loss aversion, framing, self-control, overconfidence) and say it's a **systematic** (not random) deviation.

Bridge to Lecture 5 (Ch. 6 — Demand curves & elasticities). We now have a consumer who picks the tangency bundle for *given* prices and income. **Change one price and re-solve**, over and over, and the optimal quantities trace out a **demand curve** — closing the loop back to the demand we assumed in Lecture 2, and connecting to the **elasticities** of Lecture 3.

★ Practice problems (with solutions)

A full-chapter set: preferences & axioms → utility & MU → MRS → budget line & MRT → the optimum → Cobb-Douglas shortcut → corner solutions → comparative statics → behavioral. Try each before opening the solution. Use **Food (F)** on the x-axis, **Entertainment (E)** on the y-axis throughout.

1. A consumer has bundles $A = (E\ 4, F\ 2)$, $B = (E\ 8, F\ 2)$, $C = (E\ 3, F\ 3)$, $D = (E\ 6, F\ 5)$. Using only "**more is better**," which rankings can you establish for certain, and which need the indifference map? (Compare A–B, C–D, and A–C.)
2. Preferences satisfy completeness, transitivity, monotonicity. A student says: "*I prefer coffee to tea, tea to cocoa, but cocoa to coffee.*" Which **axiom** does this violate, and why is it a problem for building a utility function?
3. $U = \sqrt{EF}$. (a) Find the utility of $(E\ 2, F\ 8)$, $(E\ 4, F\ 4)$, $(E\ 8, F\ 2)$, $(E\ 5, F\ 5)$. (b) Which lie on the same indifference curve? (c) Rank all four bundles.
4. $U = F^{0.5} E^{0.5}$. (a) Derive MU_F and MU_E . (b) Evaluate both at $(F\ 4, E\ 9)$. (c) Does MU_F rise or fall as F increases (E fixed)? Name the principle.
5. For $U = \sqrt{EF}$, show that $MRS = E/F$. Then compute the MRS at $(E\ 2, F\ 8)$ and at $(E\ 8, F\ 2)$ and interpret the difference in one sentence.
6. Income $I = 40$, $p_F = 4$, $p_E = 2$. (a) Write the budget line as $E = \dots$. (b) Give both intercepts and the slope. (c) Is $(F\ 6, E\ 8)$ affordable? (d) What is the **MRT**, and what does it mean?
7. Same as 6 ($I = 40$, $p_F = 4$, $p_E = 2$). (a) The government gives a **€20 income top-up**. What happens to the budget line? (b) Instead, p_F falls to 2. What happens? (c) Sketch the difference in words (shift vs pivot).
8. $U = \sqrt{EF}$, $I = 40$, $p_F = 4$, $p_E = 2$. Find the **optimal bundle** using $MRS = MRT$ + the budget line. Verify it satisfies the constraint.
9. $U = F^{0.25} E^{0.75}$ (Cobb-Douglas), $I = 60$, $p_F = 3$, $p_E = 6$. Use the **expenditure-share shortcut** to find F^* and E^* . What share of income goes to each good?
10. (Perfect substitutes.) $U = 2F + E$, $I = 24$, $p_F = 3$, $p_E = 2$. (a) Compute MU_F/p_F and MU_E/p_E . (b) What is the optimal bundle, and why is it a **corner solution**?
11. (Perfect complements.) $U = \min(F, E)$, $I = 30$, $p_F = 2$, $p_E = 3$. Find the optimal bundle. (Hint: the optimum sits at the corner where $F = E$; combine with the budget line.)
12. (Comparative statics.) Start from problem 8's optimum ($U = \sqrt{EF}$, $I=40$, $p_F=4$, $p_E=2$). Now p_F falls to €2. (a) Find the new optimum. (b) Did the quantity of Food rise or fall? (c) You now have **two (p_F, F^*) points** — what curve are you tracing out?
13. (Behavioral.) Name the bias in each: (a) You keep a concert ticket you'd never pay today's resale price for. (b) "1 in 3 patients survive" feels better than "2 in 3 die." (c) You put your alarm clock across the

room so you can't hit snooze. (d) 80% of new founders rate their success odds $\geq 7/10$.

Solutions

1. **A vs B:** B has more E (8 vs 4), same F $\rightarrow B > A$ (more is better). **C vs D:** D has more of *both* (E $6 > 3$, F $5 > 3$) $\rightarrow D > C$. **A vs C:** A has more E ($4 > 3$) but less F ($2 < 3$) $\rightarrow$ **cannot rank** by "more is better" alone; you need the **indifference map** (depends on the MRS).
2. Violates **transitivity**. coffee > tea and tea > cocoa should imply coffee > cocoa, but she says cocoa > coffee — the preferences **cycle**. A cycling ranking **can't be represented by a utility function** (no set of numbers can have coffee > tea > cocoa > coffee), so the whole model of choice breaks down.
3. (a) $U = \sqrt{EF}$: (E2,F8) $\rightarrow \sqrt{16} = 4$; (E4,F4) $\rightarrow \sqrt{16} = 4$; (E8,F2) $\rightarrow \sqrt{16} = 4$; (E5,F5) $\rightarrow \sqrt{25} = 5$. (b) The **first three** (all $E \cdot F = 16$) lie on the **same indifference curve** ($U=4$). (c) **(E5,F5) > {the other three, which are indifferent to each other}** — i.e. $(5,5) > (2,8) \sim (4,4) \sim (8,2)$.
4. (a) $MU_F = \partial U / \partial F = 0.5 F^{-0.5} E^{0.5} = 0.5 \sqrt{E/F}$; $MU_E = 0.5 F^{0.5} E^{-0.5} = 0.5 \sqrt{F/E}$. (b) At (F4,E9): $MU_F = 0.5 \sqrt{9/4} = 0.5 \cdot 1.5 = 0.75$; $MU_E = 0.5 \sqrt{4/9} = 0.5 \cdot 0.667 = 0.333$. (c) As F rises, $\sqrt{E/F}$ falls $\rightarrow MU_F$ falls = **diminishing marginal utility**.
5. $MRS = MU_F / MU_E = [0.5 \sqrt{E/F}] / [0.5 \sqrt{F/E}] = \sqrt{E/F} \cdot \sqrt{E/F} = E/F$. At (E2,F8): $MRS = 2/8 = 0.25$; at (E8,F2): $MRS = 8/2 = 4$. **Interpretation:** with lots of F she'll give up only 0.25 E for one more F; with little F she'll give up 4 E — the MRS **falls** as F rises (diminishing MRS / convex curve).
6. (a) $40 = 4F + 2E \rightarrow E = 20 - 2F$. (b) E-intercept = 20, F-intercept = 10, slope = -2. (c) (F6,E8): $4 \cdot 6 + 2 \cdot 8 = 24 + 16 = 40 \rightarrow$ **exactly affordable** (on the line). (d) $MRT = p_F / p_E = 4/2 = 2 \rightarrow$ giving up 1 Food frees €4, which buys 2 Entertainment.
7. (a) **Income +€20** ($I = 60$): **parallel outward shift** — new intercepts $E = 60/2 = 30$, $F = 60/4 = 15$; **slope unchanged** (-2). (b) **p_F falls to 2:** budget **pivots** — E-intercept **unchanged at 20**, F-intercept rises to $40/2 = 20$, slope flattens to $-p_F / p_E = -1$. (c) Income change = **parallel shift** (same slope, more of everything); own-price change = **pivot** (one intercept fixed, slope changes).
8. $MRS = E/F$ (from problem 5). Set $MRS = MRT$: $E/F = 4/2 = 2 \rightarrow E = 2F$. Budget: $4F + 2E = 40 \rightarrow 4F + 2(2F) = 40 \rightarrow 8F = 40 \rightarrow F^* = 5, E^* = 10$. (Check: $4 \cdot 5 + 2 \cdot 10 = 20 + 20 = 40$ ✓.)
9. Shortcut for $U = F^{0.25} E^{0.75}$ ($a=0.25, b=0.75, a+b=1$): $F^* = 0.25 \cdot 60/3 = 0.25 \cdot 20 = 5$; $E^* = 0.75 \cdot 60/6 = 0.75 \cdot 10 = 7.5$. **Shares:** 25% of income (€15) on Food, 75% (€45) on Entertainment. (Check: $3 \cdot 5 + 6 \cdot 7.5 = 15 + 45 = 60$ ✓.)
10. $U = 2F + E$: $MU_F = 2$, $MU_E = 1$. (a) $MU_F / p_F = 2/3 \approx 0.67$; $MU_E / p_E = 1/2 = 0.5$. (b) Food gives **more utility per euro** ($0.67 > 0.5$) and the rate is **constant** (perfect substitutes), so spend **everything on Food**: $F^* = I / p_F = 24/3 = 8$, $E^* = 0$. **Corner solution** because $MU_F / p_F \neq MU_E / p_E$ everywhere — there's no interior tangency; the highest indifference (straight) line touches the budget line **on the F-axis**.
11. $U = \min(F, E)$: no point in having more of one than the other, so the optimum has **$F = E$** . Budget: $2F + 3E = 30$ with $E = F \rightarrow 2F + 3F = 30 \rightarrow 5F = 30 \rightarrow F^* = E^* = 6$. (Check: $2 \cdot 6 + 3 \cdot 6 = 12 + 18 = 30$ ✓.) (Note: the price ratio is irrelevant to the ratio of goods — only to how much of the fixed bundle you can afford.)

12. Now $p_F = 2$, $p_E = 2$, $I = 40$. $MRS = E/F = p_F/p_E = 2/2 = 1 \rightarrow E = F$. Budget: $2F + 2E = 40$ with $E = F \rightarrow 4F = 40 \rightarrow F^* = 10, E^* = 10$. (b) Food rose from 5 to 10 as its price fell $4 \rightarrow 2$ (a downward-sloping demand response). (c) The two points ($p_F=4, F=5$) and ($p_F=2, F=10$) are two points on the consumer's demand curve for Food.

13. (a) **Endowment effect** / **sunk-cost + status-quo bias** (you value it more because you own it). (b) **Framing effect** (same facts, "survive" vs "die"). (c) **Lack of self-control** — a commitment device (planner constrains the future doer). (d) **Overconfidence**.

★ Fun facts & memorable details (Lecture 4)

Sticky bits from Consumer Choice.

- **"More is better" has a loophole:** economists just **redefine bads as the absence of the bad** — you don't demand pollution, you demand *clean air*. A neat trick that keeps monotonicity intact.
- **Utility is unobservable — so numbers lie.** $U = 8$ is not "twice as good" as $U = 4$; utility is purely **ordinal**. Early economists agonised over this; the modern answer is "we only ever need the ranking."
- **The bang-for-your-buck rule is the whole optimum.** $MU_F/p_F = MU_E/p_E$ — at the best bundle, the **last euro** spent on each good buys the **same** extra utility. If it didn't, you'd move money to the better-value good. That's *all* tangency means.
- **Cobb-Douglas people are creatures of habit:** with $U = F^a E^b$ you always spend the **same share** of income on each good — no matter how rich you get or how prices move. That's why it's the field's favourite functional form.
- **Left shoes are worthless alone.** Perfect complements (L-shaped curves) are why a single earring, one glove, or a lone right shoe adds **zero** utility — you need the matching unit.
- **The ultimatum game breaks "Homo economicus."** Offered a lopsided split, people **pay real money to punish unfairness** — and this holds from Pittsburgh to Papua New Guinea. Fairness is not a Western quirk.
- **You'd sell your coffee mug for far more than you'd pay for it.** The **endowment effect**: Cornell students valued a randomly-given mug **2–4× higher** than students who didn't get one. Ownership rewires valuation.
- **The paradox of voting:** a strict cost-benefit reading says *don't vote* (your vote almost never decides the election). That people vote anyway is a clue that **narrow self-interest is too small a theory of us**.
- **Ulysses tied himself to the mast** to resist the Sirens — a 3,000-year-old **commitment device**, the same logic as not keeping cigarettes in the house. Self-control problems are ancient.